

A Gossip-Based Auction Algorithm for Decentralized Task Rescheduling in Heterogeneous Drone Swarms

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One of the key technologies used in scheduling for drone swarms, which execute complex tasks and have broad application areas, is task assignment. Scheduling ensures the efficient assignment and execution of tasks in drone swarms by considering time constraints and environmental variables. The dynamic nature of the environments in which drone swarms operate requires the swarm to possess a distributed replanning capability that can dynamically accommodate unassigned or newly added tasks while adhering to time constraints. The distributed approach enables rapid task allocation without a central node, allowing drones to freely join or leave the swarm, thereby enhancing both resilience and flexibility. In this article, a distributed method called Harmony drone task allocation (DTA) is proposed for solving the multidrone task assignment problem with complex time window constraints. The proposed Harmony DTA aims to minimize total system cost and task execution delay while ensuring the conflict-free assignment of all tasks within their valid time intervals in dynamic environments with communication constraints. By utilizing a consensus-based auction mechanism and integrating a gossip-based approach, Harmony DTA efficiently minimizes communication load while ensuring that tasks are assigned to the most appropriate drones, balancing task urgency with resource availability. Simulations demonstrate that the proposed method can effectively assign newly emerging time-limited tasks among drones in dynamic environments. Results from Monte Carlo simulations show that Harmony DTA provides assignments with lower total costs and reduced total message size. In addition, the developed gossip-based consensus algorithm has been demonstrated through simulations to deliver conflict-free assignments in communication-limited environments.

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I. INTRODUCTION

Drones are a major innovation of modern technology and have been transformed into a very effective and fast solution in agriculture, logistics, security, and emergency response. Drones are advantageous compared with other aircraft because of their lower operational costs and ease of use. They are able to perform dangerous missions, and undertake remote operations in hazardous areas, without the risk to human life. The wide range of applications for which drone technology is needed clearly indicates the importance and indispensability of drone technology in the modern world.

The multiagent systems (MAS) are composed of various autonomous entities called agents which interact in an environment in order to accomplish individual or collective goals. An agent in this context is an independent unit capable of perceiving its environment, processing information, and taking actions to achieve specific objectives. MASs, such as autonomous aerial vehicle (AAV) systems, are among today's popular research topics. Compared to a single agent, MASs can perform more complex tasks and enhance execution efficiency through cooperation [1].

While individual drones excel in many tasks, increasingly complex missions necessitate systems involving multiple drones. Drone swarms, in which numerous drones act in a coordinated manner, communicate, and perform tasks jointly, address this need. These swarms are designed to tackle missions beyond the capabilities of a single drone. One major advantage of drone swarms is their resilience; each drone can function independently, so a malfunctioning drone does not compromise the overall performance of the swarm. In addition, drone swarms can operate adaptively and flexibly in dynamic and changing environments [2].

Drone swarms can consist of either homogeneous or heterogeneous aerial vehicles. Homogeneous swarms have drones of the same characteristics and capabilities, whereas heterogeneous swarms consist of drones of different types and features. Some advantages are offered by heterogeneous swarms. For example, multiple sensors and equipment make it possible to execute diverse tasks simultaneously. The flexibility and adaptivity of the swarm to environmental variability is increased. In addition, the employment of specialized drones, which enhances mission success, enables the execution of more complex and multifaceted operations.

The biggest disadvantage of drone swarms is that coordination problems are highly challenging and complex [3]. The issues being addressed in drone swarms generally involve problems with a high degree of complexity, classified as NP-hard. An NP-hard problem is a mathematical problem for which a solution can be verified, but no known algorithm runs in polynomial time to solve it, meaning that the solution time increases exponentially with the size of the problem. Therefore, finding optimal solutions for these problems is often impossible or very time-consuming [4].

Effective cooperation among drone swarms is crucial for improving mission performance. Without coordination, resources can be wasted, and task costs can increase without

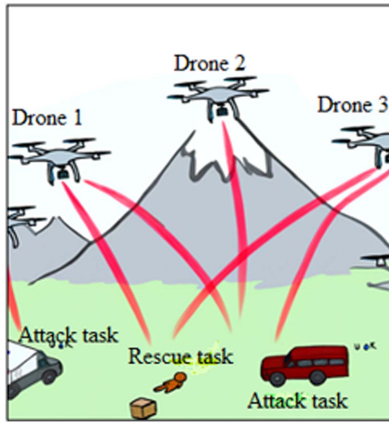


Fig. 1. Task allocation scenario for a drone swarm.

significant performance gains. Drones operating in systems with different capabilities and complex, dynamic missions are difficult to coordinate. This implies that developing algorithms for managing drone coordination in dynamic, noisy, and uncertain environments is critical. Performing complex tasks effectively and efficiently in drone swarms is a current research focus [5].

Drone swarm planning is one of the core technologies of swarm systems, determining whether and when a mission can be completed [6]. The task assignment problem in drone swarms seeks to allocate tasks among drones in a way that maximizes the overall system benefit [7]. This challenge is especially significant in heterogeneous drone swarms with varying capabilities, as shown in Fig. 1. The multidrone task reassignment problem is an NP-hard combinatorial optimization problem under complex constraints [8]. The uncertainties of dynamic environments and enemy attacks can complicate the execution of preplanned missions. Therefore, multidrone systems must be able to quickly respond to sudden events and update their task plans in real time through feedback mechanisms. The primary goal of task reassignment is to improve mission effectiveness and reduce the risk of failure after sudden changes.

Scalability is one of the key features in drone swarms, as drones can flexibly join or leave the swarm. However, as the number of tasks and drones increases, finding an optimal solution by considering every combination of drone-task assignments becomes computationally expensive [9].

However, in the context of real-world environments, it is assumed that ideal communication conditions, e.g., low latency, high reliability between drones are often assumed, though they are not always achievable. Practically, communication among drones is susceptible to bit errors, signal attenuation, and packet loss. In communication restricted environments, where there are limited drones, the problem becomes even more complex in that they must be reasonably assigned to specific tasks [10].

Moreover, in mission-critical operations, it is essential to be vigilant against the presence of traitorous agents within the swarm. These traitor agents could be compromised drones that attempt to sabotage the mission by

disseminating false information, delaying task completion, or misguiding other drones. Such malicious behavior can severely undermine the effectiveness of task assignments and jeopardize the overall mission success. Therefore, task assignment algorithms must incorporate mechanisms to detect and isolate these traitorous agents promptly, ensuring that only trustworthy drones are assigned to critical tasks. Integrating trust-based models and security protocols can help in safeguarding the swarm against these internal threats and maintaining the integrity of the mission.

Various methods have been proposed to solve the task assignment problem, mainly divided into centralized and distributed approaches. Centralized methods are a traditional approach for task planning in drone swarms. All drone and task information is collected in a single node, the best plan is generated, and the results are distributed. Although this method is successful in terms of optimization performance, the computation time rapidly increases with the number of drones and tasks, reducing the system's scalability. Moreover, if the single node fails, the entire system may collapse, creating a security vulnerability. While multicentralized approaches mitigate this issue, they cannot solve the computational explosion problem [6]. Centralized planning is far from sustainable, especially in large-scale systems, due to high communication load and bandwidth limitations. Nevertheless, intelligent algorithms, such as genetic algorithms (GA) and particle swarm optimization are commonly used in centralized methods [8].

Due to the low-cost and self-organizing nature of swarms, each drone cannot handle the computational load of centralized planning, and centralized servers face communication pressure from gathering information. Therefore, swarms rely on distributed task planning. Centralized algorithms are more susceptible to errors and change than distributed algorithms, which are more resilient and flexible, and respond to changes in situational awareness (SA) much faster. This is because task assignments are both more efficient in terms of communication as well as computation. Particularly during conflict resolution phase, local communication between drones can resolve conflicts more efficiently [10].

In large-scale systems and dynamic environments, distributed methods are more suitable than centralized approaches due to their strong fault tolerance and flexibility. Each drone can compute reassignment plans in parallel, enabling rapid responses. Distributed algorithms can be categorized into three main groups: 1) redundant centralized computation; 2) distributed parallel optimization; and 3) market-based auction methods.

Redundant centralized computation assumes perfect communication with unlimited bandwidth and distributes the centralized planner across all vehicles, eliminating single points of failure. However, inconsistencies in shared SA can lead to allocation conflicts since each vehicle optimizes tasks based on its own dataset. Distributed parallel optimization divides the global optimization problem into subproblems, improving performance and efficiency through parallel computation, though it requires high-speed

data exchange and is better suited for gigabit-connected systems.

Market-based auctions are the most widely used distributed method that prevents task conflicts by independent selection and distributed negotiation. Each AAV independently selects a task and communicates with the swarm to resolve conflicts. The method reduces communication traffic extremely, and makes system resilience better, as it guarantees consistency after the negotiation phase. Also, each AAV can compute independently, reducing the computational load and allowing heuristic methods like greedy algorithms to further reduce computation allowing for timely planning. A conflict free task plan is reached after several planning and conflict resolution iterations [6]. Consensus-based bundle algorithm (CBBA), performance impact (PI), and hybrid information and plan consensus (HIPC) are the most representative of these methods, and many approaches have been derived from them.

Agents in a distributed system interact through a network to achieve a common goal, with difficulties and mistakes inevitable. In a multiagent system, consensus is the process by which independent agents agree on a decision or action even though the agents may have individual goals or priorities. Consensus is ensured in distributed task assignment algorithms, which allows the system to function harmoniously and coordinated, thereby increasing overall performance and reliability. Fast and scalable information dissemination is achieved by gossip protocols that enhance consensus. While these protocols make it possible for each agent or node to exchange random information with its neighbors, then spread the information quickly across the network to form a reliable consensus [11]. This method is especially suited for large and dynamic networks.

In this article, an innovative algorithm named harmony drone task allocation (DTA) is presented for task allocation and replanning in drone swarms within dynamic environments where time-limited tasks exist. The dynamic task allocation problem in drone swarms is defined as a multiobjective optimization problem, and reset scenarios requiring replanning are also explained within the scope of this problem. Unlike greedy algorithms in literature, Harmony DTA distinguishes itself with its two-stage auction structure, taking into account the synergy between tasks during assignment. This increases the overall system efficiency while reducing the total message size used during task assignment, which enhances its usability in bandwidth-limited environments.

The distributed structure of the proposed algorithm, similar to other distributed task assignment algorithms in literature [5], [6], [10], [11], [12], [24], focuses on obtaining feasible and approximate solutions rather than guaranteeing extreme points. This is because the algorithm is based on local decisions and iterative updates, prioritizing real-time adaptability over exact global optimality. Therefore, the algorithm does not guarantee the existence or attainment of extreme points, but it provides practical suboptimal solutions for dynamic and nonconvex problem spaces.

The gossip-based consensus structure prevents conflicting assignments. In addition, the algorithm features a modular design that offers flexibility in the task assignment process and includes additional submodules to enhance its applicability to drone swarms. The first of these submodules monitors the drone's real-time system health level, aiming to improve the swarm's task success. Another submodule includes a trust-based mechanism to detect and isolate malicious or faulty agents within the swarm, improving task assignment accuracy and ensuring the system's security and operational integrity. Furthermore, an agent communication language was developed to improve the algorithm's applicability and performance in task assignment. While most task allocation algorithms in literature focus on static systems or systems operating with limited environmental variables, Harmony DTA not only provides a task allocation algorithm that performs well in dynamic and rapidly changing conditions but also incorporates additional components to enhance task assignment and swarm success.

The performance and applicability of the proposed algorithm for task allocation are presented with detailed simulation results. The simulations demonstrate that the Harmony DTA algorithm can generate effective solutions within limited time frames and provide near-optimal results. The Harmony DTA algorithm has been extensively compared with auction-based task allocation methods, including the CBBA, PI, and HIPC algorithms. These comparisons were conducted in environments with and without communication constraints. Harmony DTA outperformed other algorithms in terms of communication load, reducing the total message size by 15%. This success is attributed to the compact agent communication language integrated into the Harmony DTA architecture, which includes only essential information [sender/receiver IDs, task cost, unique codes (UCs)]. This structure enables effective solutions with small-sized messages, significantly reducing communication costs. In comparisons based on the optimality metric, the PI algorithm achieved the best results in all scenarios due to its focus on global optimization. However, this superior performance comes at the cost of higher communication and computational overhead. The CBBA and HIPC algorithms, relying more on local optimization strategies, lagged behind PI in terms of optimality. Harmony DTA, on the other hand, not only focused on local optimization but also incorporated an innovative structure that accounts for the synergy between tasks. This approach allowed Harmony DTA to achieve results 7% better than CBBA and HIPC, although it remained below the PI algorithm. By balancing communication and computational load while providing near-optimal solutions, Harmony DTA emerges as a more practical approach compared to the PI algorithm.

Furthermore, to demonstrate the applicability of Harmony DTA in real-world scenarios, its adaptation to the highly innovative technology of divisible drone swarms was also addressed. The limited number of studies in literature on distributed decision-making for divisible drone swarms highlights the significance of this work. The application of

Harmony DTA to this groundbreaking technology clearly demonstrates the algorithm's flexibility and potential.

The rest of this article is organized as follows: Section II reviews related work. Section III outlines the task assignment problem formulation. Section IV introduces the proposed algorithm. Section V presents simulation results. Finally, Section VI concludes this article.

II. RELATED WORK

Drone swarms have garnered significant attention in the field of autonomous systems, demonstrating considerable advantages across various domains. Academic literature consistently highlights the benefits of drone swarms, including their efficiency, scalability, autonomy, task specialization, and resilience. A survey conducted in [12] underscores the widespread application of drone swarms in areas, such as agriculture, civil infrastructure monitoring, ecological surveillance, and exploration activities. In addition, drone swarms are effectively employed for security operations [13], [14], disaster management [15], and leisure activities [16].

The task assignment problem involves deploying various agents to complete different tasks within given constraints. Gerkey and Mataric [17] introduced a domain-independent framework for classifying multitask (MT) assignment problems, categorizing key elements into single-task (ST) and MT robots, single-robot and multirobot (MR) tasks, and distinguishing between instant and timed assignments.

Skaltsis et al. [18] provided a comprehensive review of task assignment methods for AAV swarms, detailing the widespread application areas of AAVs and the different task assignment algorithms used in these domains. The study particularly focuses on the complexity, optimality, and scalability of task assignment algorithms, highlighting the differences between centralized and distributed task assignment approaches. Chakraa et al. [19] provided an in-depth analysis of optimization techniques in MT assignment problems. The study categorizes market-based, behavior-based, and optimization-based approaches used to solve MT assignment problems and discusses the advantages and limitations of these methods. It particularly emphasizes that optimization-based techniques outperform other strategies in terms of performance and computational complexity. Peng et al. [20] provided a comprehensive review of dynamic task allocation methods for AAV swarms oriented to ground targets. The article systematically summarizes the establishment and solution methods of task allocation models and discusses the advantages and disadvantages of common dynamic task allocation algorithms, such as market-based mechanisms, intelligent optimization algorithms, and clustering algorithms. The works in [21] and [22] are also key publications that thoroughly examine task assignment algorithms in AAV swarms.

The first method proposed for the centralized solution of the task assignment problem is the Hungarian algorithm, a deterministic optimization method [23]. In 1957, James

Munkres demonstrated that this algorithm operates in polynomial time, independent of cost. Consequently, the algorithm is also referred to in literature as the "Kuhn–Munkres Algorithm" or the "Munkres Algorithm." The Hungarian Algorithm guarantees finding an optimal solution in polynomial runtime, with a worst-case complexity of $O(n^3)$.

The Ant Colony Algorithm, introduced by Dorigo et al. [24], is a heuristic optimization technique originally designed to tackle complex combinatorial optimization problems, such as the traveling salesman and quadratic assignment problems. This algorithm has also been applied to task allocation in AAV swarms, with various examples found in literature. For instance, in [25], the authors explored the use of AAVs in agricultural tasks, focusing on the proximity between AAVs and tasks as well as the resources they carry. They propose the capability value sensitive-collection path ant colony optimization (CVS-CPACO) method, which yields better solutions and reduces resource wastage compared to CPACO and modCPACO, although the method's distance metric is slightly less optimal than the benchmarks. The authors in another study [26] addressed the task allocation and track planning challenges for multiple AAVs attacking ground targets. The researchers developed an ant colony optimization algorithm that features adaptive parameter adjustment and bidirectional search (BSAP-ACO) for cooperative path planning, alongside a novel particle swarm optimization algorithm based on a guidance mechanism (GMP SO) for handling the moving target problem.

The GA is a prominent method in evolutionary computation and is widely applied in AAV task allocation [27]. The number of iterations needed to generate chromosomes that meet specific conditions increases, the optimization time grows exponentially with the scale, making large-scale dynamic task allocation challenging to achieve. Unlike other heuristic algorithms, the GA employs probabilistic change rules to guide the search direction, thereby reducing the risk of converging to local optimal solutions. Most of literature on task allocation for AAV collaboration utilizes an adaptation model with a fitness function and coding before applying the GA, a process that arises from the algorithm's specific characteristics [28]. Xu et al. [29] proposed an improved GA using target bundle encoding to solve cooperative multitasking in heterogeneous AAVs. The algorithm groups tasks into goal bundles, generating individuals based on task execution order while respecting priority constraints. In [30], the task allocation problem in reconnaissance and attack missions involving multiple AAVs is addressed through an improved simulated annealing fusion GA (ISAFGA). This algorithm features enhanced solution acceptance criteria compared to traditional simulated annealing, resulting in reduced solving time and increased efficiency. Li et al. [31] explored a GA-based dynamic programming model for ST types with a small solution scale, including target redistribution for failed strikes or new targets. However, it is limited to equal AAV and target numbers, making it unsuitable for complex

assignments. Deng et al. [32] proposed a GA coding method involving multiple types of genes. They focused on collaborative MT assignment problems for heterogeneous AAVs. Simulations on medium-sized problems showed that this method outperformed random search and particle swarm optimization. However, their focus was not on minimizing total flight time, number of unfinished missions, or total distance travelled by all AAVs.

Auction algorithms were first proposed by Dimitri Bertsekas in 1979. Dimitri aimed to assign optimal tasks by putting forward the idea of cost variation during the auction. Dimitri [33] defined negotiation as rounds, and in each round, agents who aspire to the same task try to get the task by bidding on the task. The missing aspect of this study is that no function was defined for task cost calculation.

The contract net protocol (CNP), developed by Smith in 1980, is an auction-based method widely used for distributed task assignment. Various versions of CNP are also employed for task allocation in AAV swarms [34]. Liang and Kang [35] proposed a new optimal task allocation method that enhances the CNP within a multiagent system. This method enables dynamic task allocation in response to environmental changes and improves cooperation and efficiency among AAVs. Yan et al. [36] explored cooperative search and simultaneous attack by AAV groups against multiple targets. A CNP-based task allocation algorithm was proposed to maximize system efficiency, considering track cost and AAV resource constraints in real-time environments. Yan et al. [37] proposed an improved contract network algorithm for real-time AAV reconnaissance missions offering higher efficiency and fewer auction rounds than the baseline method. Chang and Jia [38] proposed an improved contract net algorithm for AAV task allocation, enhancing resource utilization and efficiency by adding time and cooperation conditions.

Consensus-based auction algorithm (CBAA), proposed by Brunet in 2008 [39], is a distributed task assignment algorithm that integrates auction and consensus processes. Task assignment occurs through iterative auction and consensus stages, with each task being assigned to a single agent, preventing multiple assignments.

The CBBA [40], evolving from the CBAA, effectively addresses the multiple assignment problem. In CBBA, each agent maintains a list of potential tasks, and the auction process operates at the task level. CBBA consists of two stages: 1) bundle construction, where agents create bundles of tasks based on capacity and priorities and 2) conflict resolution, where agent s exchange information to ensure consensus on task allocation, reducing conflicts and improving efficiency. This robust design suits complex, dynamic environments with multiple agents and tasks.

Whitbrook et al. [41] proposed another distributed method, called PI, which aims to directly optimize the overall objective. Among the market-based methods for basic task assignment, the PI algorithm is considered the most advanced. This is because the PI algorithm can directly optimize the system and is formulated under the problem of interest. This method introduces the concept

of “importance” to evaluate the contribution of each task to the local time cost of a AAV. Compared to CBBA, PI relies on heuristic optimization and reduces the completion time of tasks when solving more time-limited assignment problems. PI reduces the total travel time by minimizing the real-time impact of tasks on AAV planning. However, its excessive focus on minimizing cost impact may lead to the neglect of nearby tasks by the AAV, causing it to continuously explore outward and fail to complete nearby tasks on time [6].

The HIPC algorithm, presented by Johnson et al. [47], aims to reduce the convergence time and the number of messages in distributed task allocation algorithms by combining local plan consensus and implicit coordination approaches. HIPC enables each vehicle to plan for a subset consisting of its neighbors and perform plan consensus to resolve conflicts [47].

Lee et al. [42] proposed a resource-oriented decentralized auction algorithm for MR task allocation, where tasks are assigned based on probabilistic bids reflecting remaining resources, favoring the robot with the shortest completion time. Yang et al. [43] applied reinforcement learning to AAV task allocation, designing an expansion-based networking scheme that allows AAVs to improve allocation through autonomous learning. Experiments showed satisfactory optimization results.

III. PROBLEM DESCRIPTION

In this section, the fundamental concepts related to the time-limited task assignment problem for heterogeneous drone swarms in dynamic environments are first introduced. Then, reset scenarios for task assignment in dynamic environments are defined, and the task assignment problem is modeled as a multiobjective optimization problem with its constraints specified. Finally, the consensus problem, which is a significant aspect of task assignment in drone swarms, is described.

A. Fundamental Model Description

The symbol t_i represents task i , and a task is defined as an action that requires a specific set of skills or abilities and must be performed at a particular location. Υ represents a task list composed of a total of N_t tasks of k_t different types

$$\begin{aligned}\Upsilon &= \{t_1, t_2, \dots, t_{N_t}\} \\ &= \left\{ t_1^1, t_2^1, \dots, t_{N_1}^1, t_{N_1+1}^2, t_{N_1+2}^2, \dots, t_{N_1+N_2}^2, \dots, \right. \\ &\quad \left. t_{N_t-1}^{k_t}, t_{N_t}^{k_t} \right\}\end{aligned}\quad (1)$$

where the number above the t symbol indicates the type of skill required to execute the task, while N_1 represents the total number of tasks of type 1, and N_2 denotes the total number of tasks of type 2.

The symbol d_j represents drone j . Ψ denotes a swarm composed of a total of N_d heterogeneous drones of k_d different types

$$\Psi = \{d_1, d_2, \dots, d_{N_d}\}$$

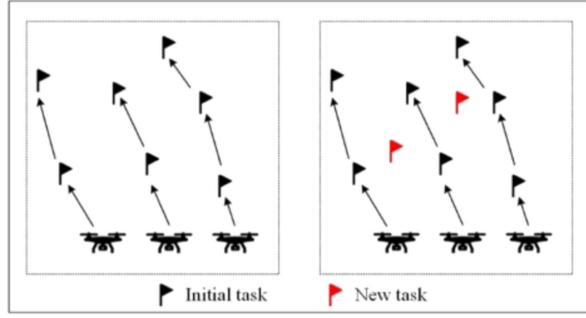


Fig. 2. Announcement of new tasks.

$$= \left\{ d_1^1, d_2^1, \dots, d_{N_1}^1, d_{N_1+1}^2, d_{N_1+2}^2, \dots, d_{N_1+N_2}^2, \dots, d_{N_d-1}^{k_d}, d_{N_d}^{k_d} \right\} \quad (2)$$

where the number above the d symbol represents the skill type of the drone, while N_1 denotes the total number of drones of type 1, and N_2 represents the total number of drones of type 2.

B. Dynamic Task Assignment Process

The MT assignment problem in heterogeneous drone swarms involves assigning N_t tasks of k_t different types to N_d drones of k_d types, with the goal of minimizing the total system cost. The concept of a “dynamic environment” refers to an operational setting that changes over time and contains uncertainties. In dynamic task assignment, after the initial tasks are assigned to drones, any newly emerging task T^* must also be assigned to the drones in the swarm.

The tasks in the full task list Υ , as shown in (3), are divided into unassigned tasks Υ_{NA} and assigned tasks Υ_A . The assigned tasks Υ_A are expressed as: $\{(\Upsilon_{A1}, \Upsilon_{I1}), (\Upsilon_{A2}, \Upsilon_{I2}), \dots, (\Upsilon_{AN_d}, \Upsilon_{IN_d})\}$. Here, Υ_{A_i} represents the tasks assigned to drone d_i . Υ_{I_i} represents the tasks assigned to drone d_i that have not yet been executed ($1 \leq i \leq N_d$)

$$\Upsilon = \Upsilon_{NA} \cup \Upsilon_A. \quad (3)$$

Once the initial tasks are assigned, each drone is given a task list, referred to as a route, denoted as P_i for drone d_i . P_i is a sequential list of tasks from the task set Υ , assigned to drone d_i . For example, if the k th element of P_i is t_j , drone d_i will perform task t_j in the k th position.

The dynamic task assignment process in drone swarms is triggered by unexpected events encountered during mission execution, requiring updates to the current task allocation. These events, such as the discovery of new tasks, task updates, or drone failures, disrupt the ongoing task plan and necessitate urgent reassessment.

In the dynamic task assignment process for drone swarms, the first reset scenario is the announcement of new tasks (Fig. 2). If a new task T^* emerges during the task assignment or execution phase, it is added to the unassigned task list

$$\Upsilon'_{NA} = \Upsilon_{NA} \cup T^*. \quad (4)$$

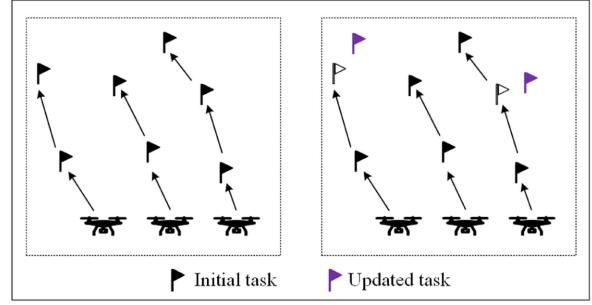


Fig. 3. Task update.

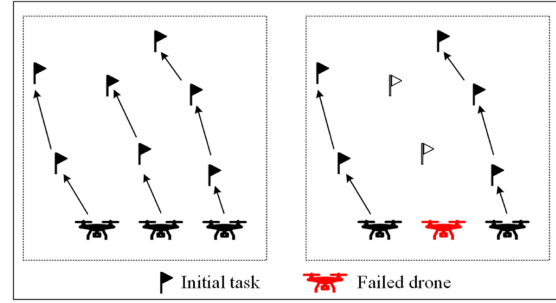


Fig. 4. Drone failure.

The second reset scenario involves task updates (Fig. 3). When significant changes occur in the location or content of tasks that have been assigned but not yet executed, those tasks are returned to the unassigned task set for reassignment. In the case where task t_j is updated to t'_j , and it was not previously assigned to a drone, it is first removed from the unassigned task list and then replaced with t'_j . However, if t_j was assigned to drone d_i , t'_j is directly added to the unassigned task list, and t_j is removed from drone d_i 's task list. If task t_j is assigned to agent d_i , then $x_{ij} = 1$; otherwise, $x_{ij} = 0$.

Update rules for the second reset scenario

$$\begin{aligned} \Upsilon'_{NA} &= (\Upsilon_{NA} \setminus t_j) \cup t'_j, & \sum_{i=1}^n x_{ij} &= 0 \\ \Upsilon'_{A_i} &= \Upsilon_{A_i} \setminus t_j \text{ and } \Upsilon'_{NA} = \Upsilon_{NA} \cup t'_j, & \sum_{i=1}^n x_{ij} &= 1. \end{aligned} \quad (5)$$

The third reset scenario involves drone failures (Fig. 4). In this case, the incomplete tasks assigned to the failed drones are added back to the unassigned task set and need to be redistributed among the remaining drones. When drone d_i fails, the list of tasks assigned to it but not yet completed is represented by Υ_{I_i}

$$\Upsilon_{I_i} = \{t_j \mid x_{ij} = 1 \text{ AND } s_j = 0\} \quad (6)$$

where s_j is a binary variable that indicates whether task t_j has been completed (1: completed, 0: not completed). For the case where drone d_i fails, the assigned and unassigned task lists are expressed by

$$\Upsilon'_A = \Upsilon_A \setminus \Upsilon_{I_i} \text{ and } \Upsilon'_{NA} = \Upsilon_{NA} \cup \Upsilon_{I_i}. \quad (7)$$

C. Problem Constraints and Optimization Objectives

In the time-constrained task assignment problem for drone swarms, it is necessary to minimize not only the total cost but also the cost arising from delayed task starts. In this study, two different optimization functions are defined: 1) total task completion cost; and 2) total time delay cost.

The task assignment problem in drone swarms is formulated as a multiobjective optimization problem in (8). In this equation, w_1 represents the cost coefficient for distance, while w_2 represents the cost coefficient for time delay

$$F = w_1 f_1 + w_2 f_2$$

$$f_1 = \min \sum_{i=1}^{N_d} \sum_{j=1}^{N_t} x_{ij} u_{ij}$$

$$f_2 = \min \sum_{i=1}^{N_d} \sum_{j=1}^{N_t} x_{ij} l_{ij}$$

$$\text{subject to: } w_1 + w_2 = 1$$

$$w_1 \geq 0, w_2 \geq 0$$

$$\sum_{i=1}^{N_d} x_{ij} = 1 \quad \text{for } 1 \leq j \leq N_t$$

$$x_{ij} \in \{0, 1\} \quad \text{for } 1 \leq i \leq N_d, 1 \leq j \leq N_t$$

$$i, j, k \in \mathbb{N}. \quad (8)$$

In (8), u_{ij} represents the distance cost incurred when drone d_i performs task t_j , and l_{ij} represents the time delay cost for drone d_i to complete task t_j , and u_{ij} is expressed in

$$u_{ij}(P_i) = \frac{\delta_{ij}}{\text{MD}} \quad (9)$$

where δ_{ij} represents the distance between drone d_i and task t_j , and MD represents the maximum possible distance between two points within the task area, which is used to normalize the cost value. δ_{ij} is calculated using the following equations:

$$\delta_{ij} = 2r \sin^{-1} \left(\sqrt{\sin^2(K) + \cos(\Phi_i) \cos(\Phi_j) \sin^2(L)} \right) \quad (10)$$

$$K = \frac{\Phi_j - \Phi_i}{2}, \quad L = \frac{\lambda_j - \lambda_i}{2} \quad (11)$$

where Φ_i represents the latitude of the drone (in radians), and Φ_j denotes the latitude of the task (in radians). Similarly, λ_i is the drone's longitude (in radians), and λ_j is the longitude of the task (in radians). The radius of the Earth, denoted as r , is approximately 6371 km.

The time delay cost l_{ij} is expressed by

$$l_{ij}(P_i) = \frac{\tau_{ij} - \tau_j^{\text{start}}}{\tau_j^{\text{final}} - \tau_j^{\text{start}}} h(t_{ij}). \quad (12)$$

$h(t_{ij})$ is a binary variable that indicates whether τ_{ij} satisfies the time window (TW) constraint. It ensures that the cost is valid within the specified TW. Outside this window, the time delay cost is assigned an infinite value, rendering it

invalid

$$h(t_{ij}) = \begin{cases} 1, & \text{if } \tau_j^{\text{start}} \leq \tau_{ij} \leq \tau_j^{\text{final}} \\ \infty, & \text{otherwise.} \end{cases} \quad (13)$$

The problem constraints are defined as follows.

Return-to-Base Constraint: All drones start their missions from the base point, execute all the tasks along their route, and then return to the base for recharging after completing their tasks. This constraint is expressed by (14). In this case, if drone d_i performs task t_k after completing task t_j , then $x_{ijk} = 1$; otherwise, $x_{ijk} = 0$. Task number zero represents the base point

$$\begin{aligned} \sum_{i=1}^{N_d} \sum_{j=1}^{N_t} x_{ij0} &\geq N_d \\ \sum_{i=1}^{N_d} \sum_{k=1}^{N_t} x_{i0k} &\geq N_d. \end{aligned} \quad (14)$$

Assignment Constraint: Each task, except for the base task, is performed once by a AAV. A AAV can be assigned multiple tasks. This constraint is expressed by

$$\begin{aligned} \sum_{i=1}^{N_d} \sum_{j=1}^{N_t} x_{ijk} &= 1 \quad \forall k \in \Upsilon \setminus \{0\} \\ \sum_{i=1}^{N_d} \sum_{k=1}^{N_t} x_{ijk} &= 1 \quad \forall j \in \Upsilon \setminus \{0\}. \end{aligned} \quad (15)$$

Capacity Constraint: A drone cannot be assigned tasks beyond its capacity. This constraint is expressed by (16). L_t represents the maximum number of tasks that can be assigned to a drone

$$\sum_{j=1}^{M_t} x_{ij} \leq L_t, \quad i = 1, 2, 3, \dots, N_d. \quad (16)$$

Time Window Constraint: Time-limited tasks must be completed within a specified TW, representing the task's lifespan. For task t_j , the TW is defined as $[\tau_j^{\text{start}}, \tau_j^{\text{final}}]$. A drone d_i must arrive at the task location within this window, at time τ_{ij} , to execute the task. This constraint is expressed by

$$\tau_j^{\text{start}} \leq \tau_{ij} \leq \tau_j^{\text{final}}. \quad (17)$$

D. Consensus Problem

Consensus problems in drone swarms arise when the agents within the swarm need to reach a common decision or agreement on a certain value or piece of information. In distributed systems, globally consistent SA is required to increase cohesion and cooperation between agents. In the distributed solution of the task assignment problem, which is the scope of this article, consensus is needed to converge to the assignment value instead of SA. Therefore, one of the problems addressed by the open auction-based task assignment process is the consensus problem.

Let us assume that the N_d agents in the swarm each have inconsistent information. Let x_i represent the information associated with drone d_i . The goal of solving the consensus

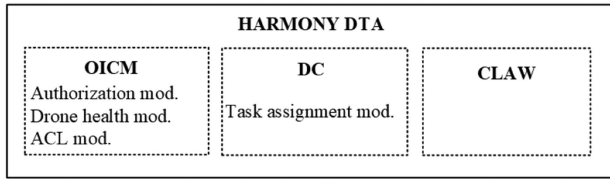


Fig. 5. Harmony DTA algorithm architecture.

problem is to have the agents share their local information with one another through communication, ultimately leading to a consensus where all agents possess the same information. The consensus problem can be mathematically expressed as follows:

$$x_i = x_j \quad \forall i, j \in \Psi. \quad (18)$$

IV. HARMONY DTA (DRONE TASK ALLOCATION) ALGORITHM

This section explains the Harmony DTA algorithm developed to solve the task assignment problem in heterogeneous drone swarms under dynamic environmental conditions. Harmony DTA is a distributed task assignment algorithm based on auctions, and it includes a gossip-based consensus algorithm to solve the consensus problem among the drones in the swarm.

The assignment of tasks in drone swarms is challenging because of the constantly changing environmental conditions, the variability of the drones, and the complexity of tasks. The Harmony DTA algorithm offers an innovative approach to address these challenges. Combining a consensus-based auction mechanism with a gossip-based protocol, Harmony DTA achieves conflict-free and efficient task assignments while minimizing communication overhead. It demonstrates high adaptability and resilience in dynamic scenarios, such as environments with limited communication, agent failures, the presence of rogue agents within the swarm, and frequent unexpected task changes. The “distributed” structure of Harmony DTA enables drones to be integrated or disintegrated from the swarm, which increases the system adaptability and robustness. Further, it minimizes the overall system cost when dividing tasks during the task allocation process and decreases communication cost, thus being efficient for networks with a limited bandwidth. These practical features place Harmony DTA in a competitive league with other task allocation algorithms.

The architecture of the Harmony DTA algorithm consists of three main modules: 1) operational integrity and communication management (OICM); 2) decision center (DC); and 3) control laws (CLAW). The main modules of the Harmony DTA architecture and their submodules are shown in Fig. 5. The OICM module is where basic-level decisions for the drone and the swarm are made. The DC module contains the core algorithms related to task assignment. The CLAW module includes the autopilot control loops that provide the necessary autonomous control for

the drone to execute the assigned tasks, but this module is beyond the scope of this study.

A. Operational Integrity and Communication Management

This module is responsible for making and executing basic-level decisions and consists of three submodules: 1) authorization; 2) drone health level control; and 3) agent communication language management. This structure aims to maximize security and operational efficiency within the swarm.

1) *Authorization Module*: The authorization module within the Harmony DTA algorithm employs a zero-trust-based authorization mechanism for drone swarms to ensure communication security. The fundamental concept is “do not trust, immediately check.” This method includes the rapid identification of authorized agents and the encryption of messages using the one-time pad (OTP) algorithm for minimal processing load and strong security. This mechanism is fundamental to the use of the UC. It is dynamic based on the drone’s ID and type and is a session specific key for encrypting and decrypting messages. The UC is calculated independently by the sender and receiver and if the decrypted UC matches, it is accepted; otherwise, the sender is flagged as unauthorized. The developed zero trust-based authorization mechanism is summarized in Algorithm 1.

2) *Drone Health Level Control Module*: This submodule evaluates the operational status of the drones. The drone’s health level is determined based on the validity of signals received from sensors or communication modules. This module evaluates the validity of signals according to weight coefficients defined based on the importance of each sensor in the system. If the drone’s health level falls below a specified threshold, it is removed from the swarm to prevent harm to the swarm.

One of the purposes of the heartbeat signal used in Harmony DTA is to detect damaged drones within the swarm. Drones send heartbeat signals to their connected drones if the health value calculated in the drone health module exceeds the threshold value. Based on whether they receive heartbeat signals from the drones to which they send their own heartbeat signals, drones determine whether their neighboring drones are damaged. If a drone does not receive an updated heartbeat signal from a neighboring drone for a duration of $5 \times \text{HTO}$ (timeout value), the neighboring drone is considered to be damaged. The damaged drone detection algorithm, executed simultaneously by all drones, is outlined in Algorithm 2.

3) *Agent Communication Language Module*: This module is a critical structure enabling drones to communicate with the swarm and ground station. It determines outgoing messages and interprets incoming ones for relevant submodules.

Algorithm 1: Harmony DTA Authorization Mechanism.

```
1: calculate  $UC_{\text{sender}}$  based on the drone's ID and type
2:  $UC_{\text{sender}} = \text{generate\_UC}(\text{drone\_ID}, \text{drone\_type})$ 
3: encrypt the message using the OTP algorithm and  $UC_{\text{sender}}$ 
4:  $\text{encrypted\_message} = \text{OTP\_encrypt}(\text{plain\_text}, UC_{\text{sender}})$ 
5: send the encrypted message along with the sender's ID and type
6:  $\text{transmit\_message}(\text{sender\_ID}, \text{sender\_type}, \text{receiver\_ID}, \text{encrypted\_message})$ 
7: receive the message at the receiver's side
8:  $\text{message\_received} = \text{receive\_message}()$ 
9: extract the sender's ID and encrypted message from the received message
10:  $\text{sender\_ID\_received} = \text{message\_received.sender\_ID}$ 
11:  $\text{sender\_type} = \text{message\_received.sender\_type}$ 
12:  $\text{encrypted\_message\_received} = \text{message\_received.encrypted\_message}$ 
13: calculate  $UC_{\text{receiver}}$  based on the received sender's ID and type
14:  $UC_{\text{receiver}} = \text{generate\_UC}(\text{sender\_ID\_received}, \text{sender\_type})$ 
15: decrypt the message using the OTP algorithm and  $UC_{\text{receiver}}$ 
16:  $\text{decrypted\_message} = \text{OTP\_decrypt}(\text{encrypted\_message\_received}, UC_{\text{receiver}})$ 
17: compare  $UC_{\text{sender}}$  with  $UC_{\text{receiver}}$ 
18: if  $UC_{\text{sender}} == UC_{\text{receiver}}$  then
19:   accept the message and process the decrypted content
20: else
21:   reject the message and flag the sender as unauthorized
22: end if
```

In the Harmony architecture, a declarative agent communication language has been developed to optimize intra-swarm coordination and task assignment. This language facilitates information exchange about tasks, supporting negotiation, auctions, and collaboration, ensuring dynamic task distribution in heterogeneous swarms.

The actions and their meanings defined for the developed communication language are presented in Table I. These actions cover commands, status information, and task updates that each agent can convey to other agents.

In Harmony ACL, each message consists of four main packets: 1) sender ID; 2) receiver ID; 3) agent type; and 4) an encoded message. The message content varies for each type of action, and the message packets for all action types defined in Harmony ACL are provided in Table II. The encoded message is obtained by encrypting the message content, which is created based on these contents, through the authorization module.

TABLE I
Actions and Their Meanings in the Developed Agent Communication Language

Action	Description
join-request	Request to join the swarm
join-approve	Approval of joining the swarm
join-rejection	Rejection of joining the swarm
leaving-request	Request to leave the swarm
leaving-approve	Approval of leaving the swarm
leaving-rejection	Rejection of leaving the swarm
liar-detect	Detection of unauthorized agent
liar-approve	Approval of unauthorized agent
pre-auction	Pre-auction request
task-announce	Task announcement
pre-auction result	Pre-auction result
synergy-auction	Synergy auction request
synergy-auction result	Synergy auction result
cd	Connection degree for gossip
rv	Ranked value for gossip
srm	Swarm Ranked Matrix
coin	Coin sending for gossip phase
reject-coin	Coin rejection for gossip phase
coin-forwarded	Coin forwarded information
gossip	Gossip for consensus
task-completed	Task completed notification
task-active	Task in progress notification

Algorithm 2: Damaged Drone Detection.

```
1: if the drone's health level  $H >$  health threshold value then
2:   Send a heartbeat signal within the heartbeat interval
3: else
4:   Do not send a heartbeat signal
5: end if
6: if a heartbeat response signal is received from drone  $d_i$  within the heartbeat timeout (HTO) then
7:   Neighbor drone  $d_i$  is in a normal state
8: else
9:   if no heartbeat signal is received from drone  $d_i$  for  $5 \times$  HTO duration then
10:    Neighbor drone  $d_i$  is in a damaged state
11:    Update the assigned and unassigned task lists
12:     $\Upsilon'_A = \Upsilon_A \setminus \Upsilon_{d_i}$ 
13:     $\Upsilon'_{NA} = \Upsilon_{NA} \cup \Upsilon_{d_i}$ 
14:   end if
15: end if
```

B. Decision Center

1) *Task Assignment Module:* The task assignment module forms the core of the Harmony DTA algorithm, where the task assignment process is executed. Harmony DTA consists of iterations between two stages: 1) a two-stage auction and 2) a gossip-based consensus phase. The auction process steps for Harmony DTA are provided in Algorithm 3.

In Harmony, after the tasks in the initial task list are assigned and the static assignment phase is completed,

TABLE II
Message Structures for Different Actions in Harmony ACL

Action	Message Structure
join-request	action type, UC
join-approve	action type, agent ID (ID of the agent newly joining the swarm), UC
join-rejection	action type, agent ID (ID of the agent rejected from joining the swarm), UC
leaving-request	action type, UC
leaving-approve	action type, agent ID (ID of the agent leaving the swarm), UC
leaving-rejection	action type, agent ID (ID of the agent whose request to leave the swarm was rejected), UC
liar-detect	action type, agent ID (ID of the unauthorized agent detected), UC
liar-approve	action type, agent ID (ID of the agent confirmed as unauthorized), UC
pre-auction	action type, task cost value, UC
pre-auction result	action type, task ID (won), maxTaskInc, UC
synergy-auction	action type, task ID (bid sent), synergy-TaskInc, UC
task-announce	action type, task ID (announced task), task location (longitude, latitude), task type, task time window, UC
synergy-auction	action type, task ID (synergy), synergy-TaskInc, UC
synergy-auction result	action type, task ID (won), synergyTaskInc, UC
cd	action type, CD, UC
rv	action type, RV, UC
srn	action type, SRM, UC
coin	action type, UC
reject-coin	action type, UC
coin-forwarded	action type, UC
gossip	action type, S_i , B_i , UC
task-completed	action type, task ID, UC
task-active	action type, task ID, UC

drones begin executing the tasks and send task completion messages for the tasks they finish. During this process, they also update their lists of uncompleted tasks using the task completion messages received from neighboring drones. Unless a reset scenario occurs, the drones continue executing the tasks in their task list and return to the base station after completing all the tasks in their route.

In the event of a reset scenario during task execution, Harmony employs a no-reset strategy. In a reset scenario, the assignments resulting from the static task allocation are preserved, and new tasks are added to the existing assignments. Since drone swarms operate in highly dynamic environments, reset scenarios can occur frequently. Using reset strategies, such as full reset in such situations would cause the task allocation process to restart continuously. Therefore, Harmony uses the no-reset approach, which provides a quick response time but may lower the quality of the solution. The Υ_{NA} list is used to control the assignment process in a reset scenario. In a reset scenario, the assignment process is managed over the Υ_{NA} list. If a reset occurs while a drone is already executing a task, it sends an active task message indicating that it will complete the

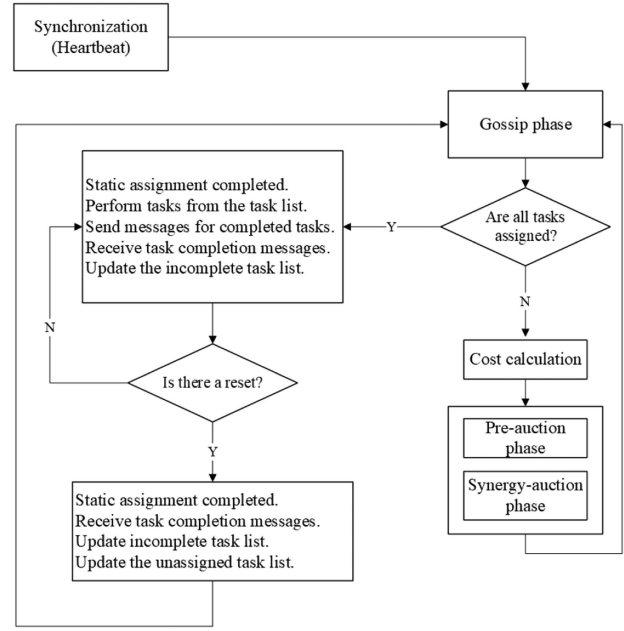


Fig. 6. Harmony DTA task assignment process.

task. Tasks for which active task messages are received are removed from the Υ_{NA} list, and no reassignment process is initiated for them. However, if a task completion message is not received for these tasks within a specified time, the reassignment process is initiated for them as well. The reassignment process for Harmony DTA is summarized in Fig. 6.

Cost Calculation: In the task assignment process, agents place auction bids based on their cost matrices. The cost matrix for drone d_i is represented as C_i and is expressed by

$$C_i = [C_{i1} \ C_{i2} \ \dots \ C_{ij} \ \dots \ C_{iM_i}] \quad (19)$$

where C_{ij} represents the cost value for drone d_i to complete task t_j . C_{ij} is calculated using (20), which is defined in Section III in line with the multiobjective optimization problem formulated to model the task multiassignment problem

$$C_{ij} = w_1 u_{ij} + w_2 l_{ij}. \quad (20)$$

u_{ij} represents the distance cost incurred by drone d_i when performing task t_j , while l_{ij} refers to the time delay cost associated with drone d_i completing task t_j .

At the beginning of each task assignment iteration, drone d_i forms its active task list A_i , which is of size N_i , based on the drone's capacity matrix Z_i and the list of unassigned tasks Υ_{NA} . $A_{ij} = 1$ indicates that task t_j is an active task for drone d_i and that the cost will be calculated for it, while $A_{ij} = 0$ indicates that task t_j is not an active task for drone d_i , and the cost should be marked as infinity

$$A_i = \{t_j \in (\Upsilon_{NA} \cap (Z_{ij} = 1))\}. \quad (21)$$

The parameter g_i indicates whether drone d_i has any elements in its active task list. $g_i = 1$ means that the drone has at least one task to bid on, while $g_i = 0$ indicates that the

drone has no tasks to bid on in the current iteration

$$g_i = \begin{cases} 1, & \text{if } \sum_{j=1}^{M_i} A_{ij} > 0 \\ 0, & \text{if } \sum_{j=1}^{M_i} A_{ij} = 0. \end{cases} \quad (22)$$

In the task assignment process, each agent maintains two cost matrices: 1) normal (C_i^N) and 2) synergy (C_i^S), and the bids are made based on these matrices. Both the normal and synergy cost matrices are calculated using (20). The normal cost matrix is the cost matrix calculated by the agent for the unassigned tasks in the relevant iteration. In the synergy cost matrix calculation, it is first assumed that the agent wins the lowest cost task from its normal cost matrix, and a new cost matrix is then calculated. If only two unassigned tasks remain, the synergy cost matrix is not calculated.

The first vector stored locally by the agents for task assignment is the task vector of size N_t . $X_i(t)$ contains the list of tasks assigned to the agent at time t . $X_{ij}(t) = 1$ indicates that drone d_i knows that task t_j has been assigned to it according to its local information at time t .

At the beginning of each auction iteration, the drone first checks whether its task list is full. If the task list is full, it does not participate in the auction and the new task assignment process

$$f_i = \begin{cases} 1, & \text{if } |P_i| < L_t \\ 0, & \text{if } |P_i| \geq L_t \end{cases} \quad (23)$$

where f_i indicates whether drone d_i will participate in the task assignment process.

Preauction Phase: It is the first phase of the auction process. In this phase, drones manage the auction process to add the lowest cost task from their normal cost matrices to their task lists. The drones broadcast a preauction message for the lowest cost task in their normal cost matrix. The selection of the task is made by identifying the task with the minimum cost, which is represented in

$$\begin{aligned} j_i^* &= \arg \min_j (C_{ij}^N) \\ \rho_i &= C_{ij_i^*}^N \end{aligned} \quad (24)$$

where j_i^* represents the index of the lowest cost task in drone d_i 's normal cost matrix, and ρ_i represents the task cost value.

After receiving auction messages from neighboring agents, the drone checks if there is another drone that has initiated an auction for the task it has also started an auction for. The variable I_i determines whether another drone has initiated an auction for the same task. If $I_i = 0$, there is no other drone; if $I_i \geq 1$, there is at least one other drone

$$I_i = \sum_{k \neq i} I(j_k^* = j_i^*). \quad (25)$$

If no other drone has initiated an auction for the same task, the agent directly adds the task to its route. If multiple drones have initiated an auction for the same task, the drone with the smallest cost value wins the task. In the case of equal bid values, the drone with the smaller ID wins the task for

the preauction phase. This entire process is formally defined and handled as described in the following equation:

Preauction result =

$$\begin{cases} \text{if } I_i = 0, & P_i \leftarrow P_i \cup \{j_i^*\} \\ & i^* = \arg \min_k (C_{kj_i^*}^N) \\ \text{if } I_i \geq 1 \text{ and } \rho_i \neq \rho_j, & P_{i^*} \leftarrow P_{i^*} \cup \{j_i^*\} \\ & \mathcal{J} = \{k \mid \rho_k = \rho_{i^*}\} \\ & i^{**} = \arg \min_{k \in \mathcal{J}} \text{ID}(k) \\ \text{if } I_i \geq 1 \text{ and } \rho_i = \rho_j, & P_{i^{**}} \leftarrow P_{i^{**}} \cup \{j_i^*\}. \end{cases} \quad (26)$$

The agent that wins the task as a result of the auction announces its victory to neighboring agents with a “*preauction result*” message. The content of the preauction result message includes the winning agent's ID, the ID of the task won, the *maxTaskInc* value, and UC. The bid value (*maxTaskInc*) for the preauction result message is calculated using the following equation:

$$\text{maxTaskInc}_i = v_i - w_i$$

$$v_i = \min_j \{R_{ij}^N\}$$

$$w_i = \min_{j \neq j_i, j \notin \text{PA}_{\text{aktif}}} \{R_{ij}^N\}. \quad (27)$$

The *maxTaskInc* value represents the additional system cost that would arise if the drone loses the task it won in the preauction phase during the synergy auction phase and must select a new task in the next round. In calculating *maxTaskInc*, the value of w is selected by excluding the tasks that have already been auctioned by other agents in the current round. The goal here is to ensure that the tasks for which an auction has been initiated are definitively assigned to a drone, preventing the drone from selecting that task in the next round. $\text{PA}_{\text{active}}$ is the list of tasks for which preauction messages have been received in the current iteration.

Synergy-Auction Phase: The phase involves an auction process where drones acquire additional tasks, known as synergy tasks, to enhance their routes at minimal cost. This auction is conducted among drones that have already won tasks during the preauction phase. Let S_i denote the number of tasks drone d_i has won in the preauction phase. Agents that did not win any tasks in the preauction phase ($S_i = 0$) are excluded from the synergy auction phase and cannot bid for synergy tasks. In addition, if a drone's task list is nearly full, with only one slot remaining ($|P_i| = L_t - 1$), it will not participate in negotiations for synergy tasks. The synergy task is selected as follows:

$$\begin{aligned} j_i^{**} &= \arg \min_j \{C_{ij}^S\} \\ \mu_i &= C_{ij_i^{**}}^S. \end{aligned} \quad (28)$$

At the end of the preauction phase, if a drone has received a preauction result message for the task with the lowest cost in its synergy cost matrix from an agent, it will send a synergy auction message (synergy-auction) for that task. The synergy auction message includes the bid value, called

TABLE III
Synergy Auction Winner Determination Rules for Drone d_i

Bid Status	Action
$\max\text{TaskInc}_i > \text{synergyTaskInc}_k$	No path change. Add the task to the winning agent and winning bid list. $S_{ij}(t) \leftarrow \text{ID}_i$, $B_{ij}(t) \leftarrow \max\text{TaskInc}_i$
$\max\text{TaskInc}_i < \text{synergyTaskInc}_k$	Remove the task from the route. Write d_k drone to the winning agent and winning bid list. $P_i \leftarrow P_i \setminus \{t_j\}$, $S_{ij}(t) \leftarrow \text{ID}_k$, $B_{ij}(t) \leftarrow \text{synergyTaskInc}_k$
$\max\text{TaskInc}_i = \text{synergyTaskInc}_k$ $\text{ID}_i < \text{ID}_k$	No path change. Add the task to the winning agent and winning bid list. $S_{ij}(t) \leftarrow \text{ID}_i$, $B_{ij}(t) \leftarrow \max\text{TaskInc}_i$
$\text{ID}_i > \text{ID}_k$	Remove the task from the route. Write d_k drone to the winning agent and winning bid list. $P_i \leftarrow P_i \setminus \{t_j\}$, $S_{ij}(t) \leftarrow \text{ID}_k$, $B_{ij}(t) \leftarrow \text{synergyTaskInc}_k$

TABLE IV
Synergy Auction Winner Determination Rules for Drone d_k

Bid Status	Action
$\max\text{TaskInc}_i > \text{synergyTaskInc}_k$	No route update. Write drone d_i to the winning agent and winning bid list. $S_{kj}(t) \leftarrow \text{ID}_i$, $B_{kj}(t) \leftarrow \max\text{TaskInc}_i$
$\max\text{TaskInc}_i < \text{synergyTaskInc}_k$	Add the task to the route. Write the task to the winning agent and winning bid list. $P_i \leftarrow P_i \cup \{t_j\}$, $S_{kj}(t) \leftarrow \text{ID}_k$, $B_{kj}(t) \leftarrow \text{synergyTaskInc}_k$
$\max\text{TaskInc}_i = \text{synergyTaskInc}_k$ $\text{ID}_i < \text{ID}_k$	No route update. Write drone d_i to the winning agent and winning bid list. $S_{kj}(t) \leftarrow \text{ID}_i$, $B_{kj}(t) \leftarrow \max\text{TaskInc}_i$
$\text{ID}_i > \text{ID}_k$	Add the task to the route. Write the task to the winning agent and winning bid list. $P_i \leftarrow P_i \cup \{t_j\}$, $S_{kj}(t) \leftarrow \text{ID}_k$, $B_{kj}(t) \leftarrow \text{synergyTaskInc}_k$

the synergyTaskInc, which is calculated using the following equation:

$$\begin{aligned} \text{synergyTaskInc}_i &= \varsigma_i - \sigma_i \\ \varsigma_i &= \min_j \{R_{ij}^S\} \\ \sigma_i &= \min_{j \neq j_i} \{R_{ij}^S\}. \end{aligned} \quad (29)$$

Let us assume that drone d_i won task t_j during the preauction phase and communicated this to the other agents in the swarm via a preauction result message. In response to this message, one or more drones sent synergy-auction bids. Suppose the highest bid was received from drone d_k . If drone d_i , the preauction winner of task t_j , receives a synergy-auction message from drone d_k (where $\forall j_k^{**} = j_i^*$), the rule set used to determine the winner is provided in Table III.

If drone d_k , which sent a synergy-auction message for task t_j , receives synergy-auction messages from other agents for the same task and finds that its own synergy value is lower than any of the other bids, it will lose the synergy task. If its synergy bid is the highest, the rule set applied to determine the winner is provided in Table IV.

Gossip and Synchronization Phase: The gossip and synchronization phase of the Harmony DTA algorithm is responsible for handling conflicts and making decisions regarding the division of tasks among the members of the swarm. This phase aims at ensuring that all agents reach a consensus on the set tasks and bids through a gossip-based consensus.

This phase follows a series of well-defined steps to synchronize the agents' decisions. Here is an overview of the key steps.

Step 1: Drones periodically send heartbeat signals to their neighbors, confirming activity. Using these signals,

each drone calculates its connection degree (CD), which is shared with its neighbors via cd messages.

Step 2: Each drone calculates its ranked value (RV) by comparing its CD with neighboring drones, using the formula

$$\text{RV}_i = \text{CD}_i \sum_{j \in V \mid (i,j) \in E} \text{CD}_j. \quad (30)$$

The RV is then shared with neighbors using rv messages.

Step 3: Drones update a swarm rank matrix (SRM) based on received RV values. This matrix is iteratively updated through srm messages until the swarm reaches consensus on the SRM.

Step 4: The drone with the highest RV value initiates graph reduction by sending a coin message to its neighbors. Drones forward coins to neighbors with lower RVs. Once completed, drones send a coin-forwarded message back, finalizing the graph reduction.

Step 5: In the ranked gossip phase, drones exchange local task assignment data with neighbors, progressing upward through the reduced graph until the top-level drone consolidates the final task assignments.

Step 6: The top-level drone then starts reverse gossip, sending the final consensus back down the graph. Each drone updates its local state, ensuring all drones agree on the final task assignments.

During the consensus phase of the assignment process, agents maintain two matrices of size N_t to reach an agreement on the current assignment: 1) the winning agent list $S_i(t)$; and 2) the winning bid list $B_i(t)$.

$S_i(t)$ is the winning agent list formed by drone d_i based on its local information for all tasks at time t . The value of $S_{ij}(t)$ is the ID of the agent that drone d_i considers the

Algorithm 3: Harmony DTA Auction Process.

```

1:  $\forall i \in \{1, 2, \dots, N_d\}$ 
2: while  $g_i = 1$  and  $f_i = 1$  do
3:   calculate  $C_i^N$  and  $C_i^S$ 
4:    $j_i^* = \arg \min_j \{C_{ij}^N\}$ 
5:    $\rho_i = C_{ij_i^*}^N$ 
6:   send preauction message
7:    $I_i = \sum_{k \neq i} I(j_k^* = j_i^*)$ 
8:   find winner for task  $j_i^*$  according to decision
     rules
9:   if the agent wins the auction then
10:     $P_i \leftarrow P_i \cup \{j_i^*\}$ 
11:     $\text{maxTaskInc}_i = v_i - w_i$ 
12:    send preauction result message
13:   else
14:     break
15:   end if
16:   receive preauction result message
17:   if  $S_i \neq 0$  then
18:     $j_i^{**} = \arg \min_j \{C_{ij}^S\}$ 
19:    if received preauction result message for  $j_i^{**}$ 
       then
20:       $\mu_i = C_{ij_i^{**}}^S$ 
21:       $\text{synergyTaskInc}_i = \zeta_i - w_i$ 
22:      send synergy-auction result message
23:    end if
24:   end if
25: end while

```

winner of task t_j in its local view. The value of $B_{ij}(t)$ is the bid value of the winning bid for task t_j according to drone d_i 's local view. At the beginning of the task assignment process, the agent initializes all its vectors to zero vectors. The update rules employed in CBBA [40] have been adapted for Harmony DTA to update these matrices.

When agent d_i receives a message from agent d_k , there are three possible actions d_i can take regarding task t_j :

- 1) *Update*: $B_{ij} = B_{kj}$, $S_{ij} = S_{kj}$;
- 2) *Reset*: $B_{ij} = 0$, $S_{ij} = \emptyset$;
- 3) *Keep*: $B_{ij} = B_{ij}$, $S_{ij} = S_{ij}$.

The rules for updating the information matrices during the gossip phase are provided in Table V.

Harmony DTA leverages heartbeat signals and gossip-based consensus mechanisms to effectively handle intermittent connections. By relying solely on local communication between neighboring agents, it minimizes the impact of connection disruptions. The algorithm optimizes communication load through graph simplification, enabling task allocation with minimal messaging. In addition, it adapts dynamically by excluding disconnected agents and reallocating tasks without halting the system. Its decentralized nature ensures resilience, as the failure of a single agent or node does not compromise the entire system. These features make Harmony DTA a robust and flexible solution for task allocation under intermittent connectivity.

TABLE V
For Drone d_i , the Gossip-Based Action Rule With Drone d_k
Regarding Task t_j

Sender Drone (S_{kj})	Receiver Drone (S_{ij})	Receiver Action
k	i	If $B_{kj} > B_{ij} \rightarrow$ Update
k	-	Update
$m \notin \{i, k\}$	-	If $B_{mj} > B_{ij} \rightarrow$ Update
$m \notin \{i, k\}$	i	Update
i	i	Keep
i	k	Reset
i	$m \notin \{i, k\}$	Reset
k	i	Reset
k	$m \notin \{i, k\}$	If $B_{kj} > B_{ij} \rightarrow$ Update
k	$n \notin \{i, k, m\}$	If $B_{kj} > B_{ij} \rightarrow$ Update
$m \notin \{i, k\}$	-	Keep
k	-	Update
$m \notin \{i, k\}$	i	If $B_{mj} > B_{ij} \rightarrow$ Update
-	i	Keep
-	k	Update
-	$m \notin \{i, k\}$	Update

C. Convergency and Computational Complexity

This section examines the convergence characteristics and computational complexity of the Harmony DTA algorithm.

PROPOSITION 1 In a fixed network structure, the Harmony DTA algorithm converges to a solution in at most $(D + 1)N_t$ iterations, where the network diameter (D) is defined as the communication distance between the farthest agents in the network, mathematically expressed as

$$D = \max_{i,k} \{\delta_{ik}\} \quad (31)$$

where δ_{ik} represents the shortest communication path between agents i and k .

PROOF The worst-case scenario occurs when all agents compete for the same lowest cost task at each iteration, and achieving consensus for assigning this task requires the maximum number of consensus steps, implying the network cannot be simplified. In such a case, the Harmony DTA algorithm operates similarly to a sequential greedy selection algorithm, disregarding synergies between tasks.

In the worst case:

- 1) only one task can be assigned per iteration. This requires N_t iterations until all N_t tasks are completed;
- 2) consensus for each task takes D iterations, as the gossip algorithm requires as many iterations as the network diameter.

For an ST, the assignment time is one iteration, and the gossip time is D iterations. Therefore, the total time for an ST is $D + 1$ iterations. For all N_t tasks, this time is

$$T_C = (D + 1)N_t. \quad (32)$$

PROPOSITION 2 The total computational complexity for Harmony DTA is expressed as follows:

$$O\left(\sum_{\alpha=1}^{k_t} N_{d\alpha} N_{t\alpha} \log N_{t\alpha} \sigma + DN_d^2\right) \quad (33)$$

where $N_{d\alpha}$ represents the number of drones of type α , $N_{t\alpha}$ denotes the number of tasks of type α , σ refers to the computational complexity of calculating the cost for an ST, and N_d signifies the total number of drones in the system.

PROOF N_{t1} is defined as the number of type-1 tasks that a type-1 drone can perform, and the total number of tasks for k_t different types is

$$N_t = N_{t1} + N_{t2} + \dots + N_{tk_t}. \quad (34)$$

The computational complexity for a type- α agent to calculate the cost for tasks suitable to its type and find the minimum value is

$$O(N_{t\alpha} \log N_{t\alpha} \sigma). \quad (35)$$

The total computational complexity for all agents to calculate the costs and find the minimum value is

$$O\left(\sum_{\alpha=1}^{k_t} N_{d\alpha} N_{t\alpha} \log N_{t\alpha} \sigma\right) \quad (36)$$

where the term σ represents the complexity of computing the cost for an ST. More specifically, it asymptotically indicates the number of operations required to execute the function used to compute the time cost of a task for an agent.

The computational complexity for the synergy task computation is also expressed as $O(N_{t\alpha} - 1) \approx O(N_{t\alpha})$, under the assumption that the values are approximately equal, which results in (35).

The computational complexity for the gossip phase depends on the network size and, in the worst case, requires iterations equal to the network diameter. It is expressed as follows:

$$O(DN_d^2). \quad (37)$$

In light of these explanations, the total computational complexity for the Harmony algorithm is expressed as shown in (33). If N_d is large, the term N_d^2 , which grows with the square of the number of agents, dominates the other terms. However, if N_d is small, the other terms become significant and affect the actual runtime of the algorithm.

An increase in the number of tasks affects the complexity linearly. However, the term dependent on the square of the number of agents becomes dominant, leading to a faster increase. Therefore, the effects of increases in both the number of tasks and agents should be evaluated together.

V. NUMERICAL RESULTS

The proposed Harmony DTA algorithm has been rigorously evaluated under various scenarios to demonstrate its effectiveness in dynamic and communication-constrained environments. This section presents simulation results, focusing on key metrics, such as optimality, reliability, communication load, and efficiency. Monte Carlo (MC) simulations were conducted to compare the performance of Harmony DTA with benchmark algorithms, such as CBBA, PI, and HIPC. These comparisons highlight the ability of Harmony DTA to achieve conflict-free task assignments

while significantly reducing communication costs. The results also underscore its adaptability and robustness in handling dynamic operational conditions.

A. Evaluation Metrics

Using various metrics to evaluate the performance of task assignment algorithms is crucial for understanding their efficiency, communication cost, and reliability. In this study, six main metrics, namely, optimality, reliability, communication load, efficiency, task effectiveness, and task feasibility, are defined to measure different aspects of the task assignment process.

1) *Optimality Metric*: The optimality metric represents the total route lengths generated for all drones after the task allocation process is completed. This metric is expressed by (38), where a smaller value indicates a more optimal result

$$M_{\text{opt}} = \sum_{i=1}^{N_d} \sum_{j=1}^{N_t} \delta_{ij}. \quad (38)$$

2) *Reliability Metric*: The reliability metric is defined in (40) as the ratio of the total number of conflicting assignments to the total number of simulations. The variable cnf_j indicates whether there is a conflict assignment for task t_j . It takes the value 1 if task t_j has a conflict assignment, and 0 otherwise

$$\text{cnf}_j = \begin{cases} 1, & \text{if } \sum_{i=1}^{N_d} x_{ij} > 1 \\ 0, & \text{otherwise.} \end{cases} \quad (39)$$

Based on this definition, the reliability metric M_{rel} is expressed as follows:

$$M_{\text{rel}} = \frac{\mathcal{F}(\text{cnf})}{N_{\text{sim}}} \quad (40)$$

where $\mathcal{F}(\text{cnf})$ represents the total number of conflicting assignments observed during MC simulations, and N_{sim} denotes the total number of MC simulations performed.

3) *Communication Load Metric*: The communication load, as a metric, represents the number of bits used in communication to solve the problem. Using the number of sent packets and their bit lengths, the communication load metric is defined as follows:

$$M_{\text{com}} = \sum_{i=1}^{N_d} \sum_p N_{\text{pkg},i,p} \cdot L_{\phi,p} \quad (41)$$

where $N_{\text{pkg},i,p}$ represents the number of type p message packets sent by AAV d_i , and $L_{\phi,p}$ denotes the bit length of a type p message packet.

4) *Efficiency Metric*: The efficiency metric considers the time each AAV spends executing the task allocation algorithm and is measured based on the AAV with the longest execution time in the system. It is mathematically expressed as follows:

$$M_{\text{eff}} = \max_{i \in \Psi} \Delta_i \quad (42)$$

where Δ_i represents the time spent by AAV d_i while executing the algorithm.

5) *Task Effectiveness Metric*: The task effectiveness of drone d_i performing task t_j is expressed as

$$E_{ij} = (TE_j - \phi \delta_{ij}) e^{-\sigma(dt_{ij})} \quad (43)$$

where:

- 1) TE_j : defined effectiveness value of task t_j ;
- 2) $dt_{ij} = \frac{\delta_{(j-1)j}}{v_i}$: time required for drone d_i to travel the distance between its previous task and task t_j (in seconds);
- 3) ϕ and σ : coefficients representing, respectively, how much the route cost and the time constraint affect task effectiveness.

The task effectiveness value E_{d_i} of drone d_i for all tasks added to its route as a result of the assignment is defined as

$$E_{d_i} = \sum_{j=1}^{|P_i|} E_{ij}. \quad (44)$$

The total effectiveness of all drones in the swarm is defined as the task effectiveness metric M_{act}

$$M_{act} = \sum_{i=1}^{N_d} E_{d_i}, \quad i = 1, \dots, N_d. \quad (45)$$

6) *Task Feasibility Metric*: The task feasibility metric represents the percentage of tasks successfully assigned while satisfying time and other constraints. It is mathematically expressed as

$$M_{feas} = \frac{N_{assigned}}{N_t} \times 100 \quad (46)$$

where $N_{assigned}$ is the number of successfully assigned tasks, and N_t is the total number of tasks. A high value indicates the algorithm's effectiveness in assigning tasks under constraints. Factors, such as task locations, TWs, and drone capacity limits may prevent all tasks from being assigned, making this metric a critical indicator of overall system performance.

B. Simulation Environment

In the simulations, tasks and drones are randomly positioned within a 4000×4000 -m area. The initial positions of all drones are considered as their base points.

Unless otherwise specified in the subsections, the heterogeneous drone swarm in the simulations consists of two types of drones: 1) attack and 2) reconnaissance. If the total number of drones N_d is even, half of the drones are designated as attack drones, and the other half as reconnaissance drones. If N_d is odd, the integer part of $N_d/2$ is designated as attack drones, and the remaining drones are reconnaissance drones. The parameters related to drone types are provided in Table VI.

In the simulations, tasks are also categorized into two types: 1) attack and 2) reconnaissance. The parameters related to task types are provided in Table VII. Attack-type drones can only execute attack-type tasks, while reconnaissance-type drones can only execute

TABLE VI
Simulation Drone Parameters

Drone Type	Speed (m/s)
Attack	10
Reconnaissance	20

TABLE VII
Task Type and Parameters

Task type	Task time window length (s)	Task duration (s)	Task effectiveness value (TE_j)
Attack	10	5	1500
Reconnaissance	15	10	1000

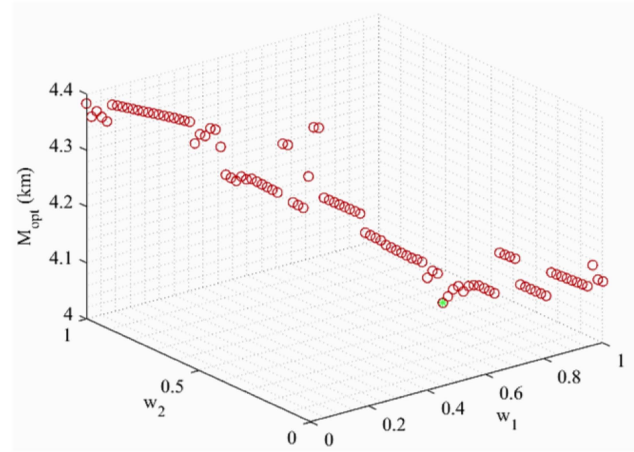


Fig. 7. Pareto-optimal task assignment distribution.

reconnaissance-type tasks. In this table, the task TW length represents the time interval during which the task is defined and must be completed. The task duration refers to the time required for the drone to complete the task, while the task effectiveness value is the value used in the calculation of the task effectiveness metric.

The simulation studies were conducted in the MATLAB environment on a computer equipped with a 3.2GHz, 8-core Intel Core i5 processor, 16GB of RAM.

C. Determination of Cost Matrix Coefficients

The weight coefficients in this part of the simulation study were determined by finding the coefficient values that minimize the total cost (M_{opt}). In this process, we increased w_1 from 0 to 1 in steps of 0.01. This was explored in the case of 20 tasks to 20 agents and with the constraint $w_1 + w_2 = 1$. The total cost of the Pareto optimal task assignment obtained through the algorithm using Harmony DTA is shown in Fig. 7, as a 3-D spatial distribution. The variations of total cost in different combinations of task distance cost and time delay cost weights are shown in this visualization.

As the magnitude of the distance cost weight w_1 diminishes from 1 to 0, the algorithm gradually reduces and subsequently increases the Pareto optimal task allocation cost. Notably, when the values of w_1 and w_2 converge, fluctuations in cost become apparent. The green circle in

TABLE VIII
Validation Simulation Results of Harmony DTA for a Static Environment

N_d	N_t	M_{feas} (%)	M_{eff} (s)	M_{rel} (%)
5	5	100	0.98	0
	10	100	1.11	0
	15	94.12	1.24	0
	20	91.15	1.31	0
10	10	100	0.99	0
	20	98.54	1.31	0
	30	93.10	1.44	0
	40	90.56	1.89	0
15	15	100	1.14	0
	30	96.12	1.34	0
	45	95.11	1.41	0
	60	93.55	1.55	0
30	30	100	1.21	0
	60	95.50	1.38	0
	90	93.47	1.64	0
	120	90.15	1.95	0

Fig. 7 indicates the specific point at which $w_1 = 0.69$ and $w_2 = 0.31$, at which the total cost has its minimum value, 4.05 km.

D. Validation Simulations

This section presents the performance results obtained through validation simulations conducted in static environments under the assumptions of a full network communication topology and lossless communication for the Harmony DTA algorithm. The validation simulations aim to demonstrate the validity of the algorithm by examining its task allocation effectiveness and efficiency under different scenarios. The simulations were conducted for environments with varying numbers of drones and tasks, and the results were evaluated using the metrics of feasible tasks (M_{feas}), efficiency (M_{eff}), and reliability (M_{rel}). In this way, the adaptability to tasks under different conditions, processing time, and reliability performance of the Harmony DTA algorithm were analyzed in detail.

The scenarios and results of the validation simulations conducted for static environments are provided in Table VIII. For each scenario, 100 MC simulations were run, and the average values obtained are presented in the table.

According to the data in Table VIII, the Harmony DTA algorithm demonstrates high feasibility under various numbers of drones and tasks. The random nature of task locations and TWs can lead to a decrease in the feasibility rate. For instance, while 100% feasibility is achieved with five drones and five tasks, this rate decreases to 90.15% as the number of drones and tasks increases. The average feasibility is calculated as 95.5%, reflecting the algorithm's adaptability. However, the impact of task location and time constraints on the feasibility rate should be considered.

The efficiency metric (M_{eff}) increases with the number of drones and tasks. While the processing time is 0.98 s for five drones and five tasks, it rises to 1.95 s for 30 drones and 120 tasks. The average processing time is calculated as 1.4 s, indicating that the algorithm is both fast and scalable.

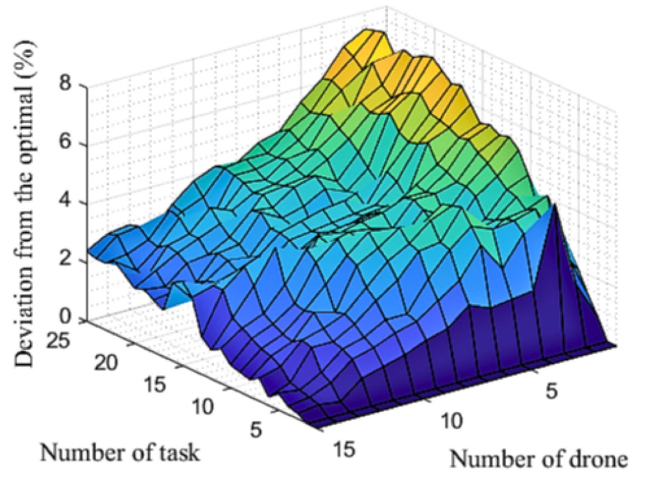


Fig. 8. Evaluating Harmony DTA algorithm performance against optimal solutions for varying agents and tasks.

The reliability metric (M_{rel}) is zero across all scenarios, proving that the algorithm does not perform conflict or erroneous assignments. These results demonstrate that the Harmony DTA algorithm is reliable and flexible, adapting effectively despite the random nature of task locations and TWs.

E. Optimality Analysis

This section analyzes how close the solutions provided by the Harmony DTA algorithm are to the optimal solution using integer programming. The task assignment performance of Harmony DTA was tested with varying numbers of agents and tasks and compared with the optimal solution. We performed analyses assuming the drones were connected with a complete graph without message loss. The results are summarized in Fig. 8. In the figure, the x -axis and y -axis represent the number of drones and missions, respectively, while the z -axis shows the percentage deviation of the objective value from the optimal solution. This graph is based on 150 MC simulations performed at each point, and the average values obtained were used. The results show that deviation from the optimal solution is, at most, about 8%.

F. Scalability Analysis

In dynamic and heterogeneous environments, such as drone swarms, scalability analysis is a critical step in evaluating the effectiveness and applicability of an algorithm. In the convergence and computational complexity section, it was explained that the proposed algorithm converges to a solution within a finite and predefined upper iteration limit.

This section focuses on the problem-solving times of the Harmony DTA algorithm, specifically the number of iterations required for various combinations of drones and tasks. Fig. 9 illustrates the convergence times for different configurations. The results show that as the number of tasks increases and the number of drones decreases, the required number of iterations generally increases. This indicates that completing more tasks with fewer drones demands more

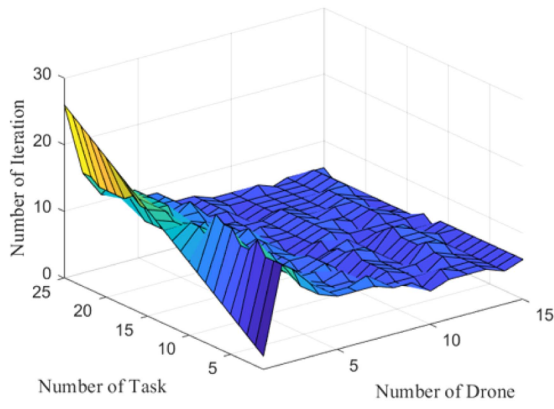


Fig. 9. Evaluating Harmony DTA algorithm converge time for varying agents and tasks.

iterations. Conversely, when the number of tasks is low and the number of drones is high, the algorithm converges in fewer iterations.

G. Analysis of the Impact of Network Structure and Imperfect Communication Conditions

Task allocation algorithms have a critical impact on system efficiency and effectiveness. Different communication network structures and scenarios where message packets are partially delivered or not delivered at all directly affect the performance of these algorithms.

In this section, simulation results conducted on four different network structures (fully connected, ring, line and star), where all messages are transmitted without any issues, are first shared to examine the impact of network structure on the performance of the Harmony DTA algorithm. Subsequently, to evaluate the effect of communication quality on task allocation performance, two scenarios are considered under a fully connected network structure: the packet loss factor alone and the combined evaluation of packet loss and bit error rate (BER) factors. Before delving into the analysis results in the first subsection, the communication model used is explained.

1) *Communication Model*: The communication model established to examine the impact of communication quality on assignment performance is presented in this section. We use the Rayleigh fading model to simulate packet loss. The Rayleigh fading model is a probabilistic model for signal degradation due to multipath propagation in a wireless communication system. Compared to simple random loss models, such as Bernoulli or Gilbert–Elliot, it provides more realistic results.

The Rayleigh model is used to measure communication quality by determining the signal-to-noise ratio (SNR) between drones and is expressed as follows [45]:

$$\gamma_{dB} = P_T + K_{dB} - P_{L_0} - 10\delta \log\left(\frac{d}{d_0}\right) - N(\mu_{dB}, \sigma_{dB}^2) \quad (47)$$

where γ_{dB} represents the SNR, P_T denotes the transmission power, P_{L_0} refers to the path loss at the reference distance d_0 ,

TABLE IX
Communication Model Parameters

Parameter	Value
P_{L_0}	80 dB
P_T	30 dBm
d_0	100 m
K_{dB}	30 dBm
μ_{dB}	-70 dBm
σ_{dB}	7 dBm
δ	2–6 (default = 2)
$\bar{P}$	5 dB

δ is the path loss exponent (ranging from 2 to 6, where $\delta = 2$ represents free space conditions, and $\delta = 6$ represents dense environments with more obstacles and signal attenuation), $N(\mu_{dB}, \sigma_{dB}^2)$ indicates the noise term expressed with mean and variance values, and K_{dB} represents the equipment gain.

Using the SNR expression defined in (47), packet loss is formulated as follows [45]:

$$\text{Packet status} = \begin{cases} \text{Packet received,} & \gamma_{dB} \geq P_{bar} \\ \text{Packet lost,} & \gamma_{dB} < P_{bar} \end{cases} \quad (48)$$

where P_{bar} represents the minimum signal power detectable by the receiver.

P_b is referred to as the BER and represents the ratio of incorrectly transmitted bits in a data stream [46]. BER is expressed as

$$P_b = 0.2 \cdot \exp\left\{-\frac{1.5}{M-1} \cdot \gamma_{dB}\right\} \quad (49)$$

where M represents the modulation level. In drone task assignments, QPSK ($M = 4$) or 16-QAM ($M = 16$) is typically preferred.

The parameter values used in the simulations for the defined communication model are provided in Table IX [10], [45].

2) *Impact of Network Structure on Assignment Performance*: This section evaluates how the assignment performance of the Harmony DTA algorithm is influenced by different communication topologies in a scenario involving ten drones and 30 tasks. The evaluation is conducted based on metrics, such as efficiency, communication load, and task effectiveness, and compared with the CBBA, PI, and HIPC algorithms. Results for each algorithm type under each topology structure are obtained through 100 MC simulations.

Fig. 10 compares the simulation results of how communication topology impacts the runtime of the Harmony DTA algorithm within the scope of the efficiency metric, alongside CBBA, PI, and HIPC algorithms. The Harmony algorithm demonstrates lower average runtime compared to CBBA and PI, with performance similar to HIPC. Despite HIPC achieving shorter minimum runtimes, Harmony offers more balanced and lower variance performance, highlighting its adaptability to various network structures and its ability to deliver consistent results.

In Fig. 11, the simulation results examine the impact of different network topologies on assignment performance

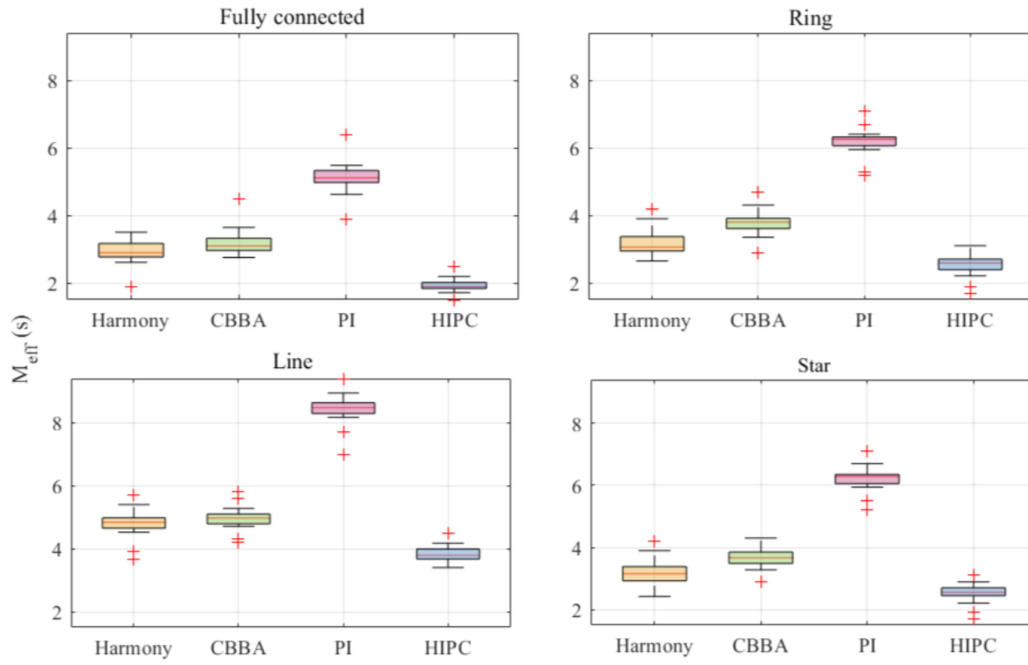


Fig. 10. Impact of communication topology on algorithm runtime (efficiency metric).

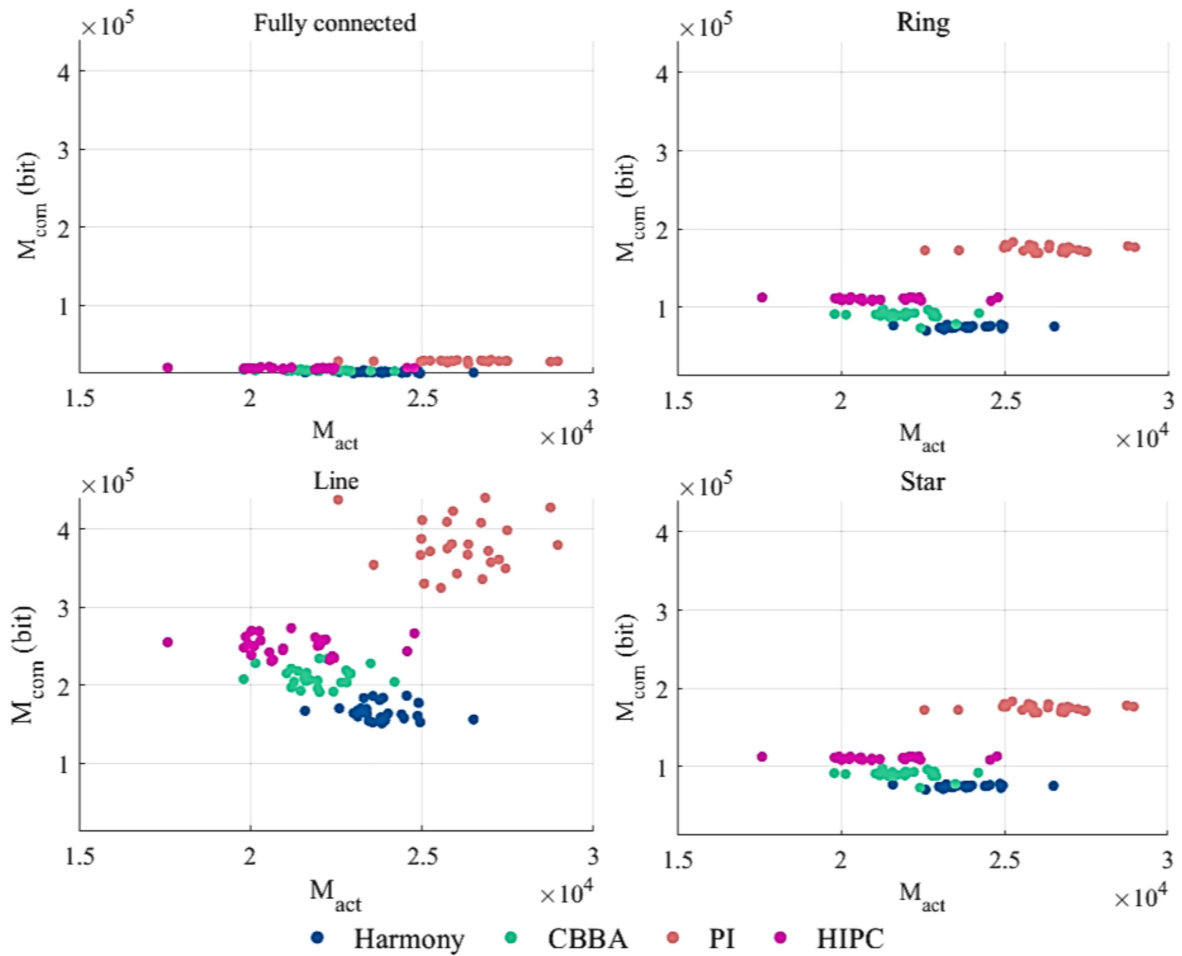


Fig. 11. Impact of communication topology on assignment performance (effectiveness and communication load metrics).

based on task effectiveness and communication load metrics. When the results are analyzed in terms of the task efficiency metric, it is observed that the network structure has no effect on the task efficiency value in any of the algorithms. However, when the results are analyzed in terms of the communication metric, it is observed that the network structure significantly affects the communication load metric, which represents the total number of bits used to solve the problem. In all algorithms, the problem is solved with a minimum communication load in a fully connected network topology. While ring topology and star topology have a similar communication load, the communication load is at its maximum in the line topology. This increase is due to the need for multiple communications to establish consensus among drones when the network is not fully connected. The results demonstrate that Harmony DTA can adapt flexibly to different topology configurations. The algorithm shows efficient performance by keeping communication and activity loads low in scenarios with low resource requirements while also being capable of reducing communication load in cases with higher requirements.

3) *Examination of Assignment Performance in Communication-Constrained Environments:* Interdrone communication is often assumed to be ideal, but real-world missions face issues like bit errors, signal loss, and packet loss. These constraints pose challenges for effective task allocation among limited drones and targets.

In this section, the impact of message packet loss on the assignment performance of the Harmony DTA algorithm is first analyzed based on the number of tasks assigned without conflicts and the efficiency metric (algorithm runtime). The analysis is conducted in a scenario with ten drones and 30 tasks.

By keeping other communication model parameters constant, the δ value is varied between 2 and 5.5 in increments of 0.5. The performance in terms of the number of tasks assigned without conflicts and the runtime of Harmony DTA are shown in Figs. 12 and 13, respectively. For each δ value, 100 simulations are performed, and the results are visualized as box plots, with average values indicated.

From Fig. 12, it is observed that as the δ value increases, the number of tasks assigned without conflicts decreases due to increased packet loss. From Fig. 13, it is seen that an increase in δ results in a slight increase in the algorithm runtime.

As the second part of this section, simulation results are presented to examine the effect of the BER on the assignment performance of Harmony DTA in imperfect communication scenarios. To analyze the effect of BER, the δ value was varied between 2 and 5.5 in increments of 0.5, and for each δ value, 100 simulations were conducted for BER values of 10^{-1} , 10^{-2} , and 10^{-3} . The average results obtained from 100 simulations for the number of tasks assigned without conflict and the efficiency metric (execution time) are shown in Figs. 14 and 15. As expected, for the same δ value, an increase in the BER value reduced the number of tasks assigned without conflict and led to an

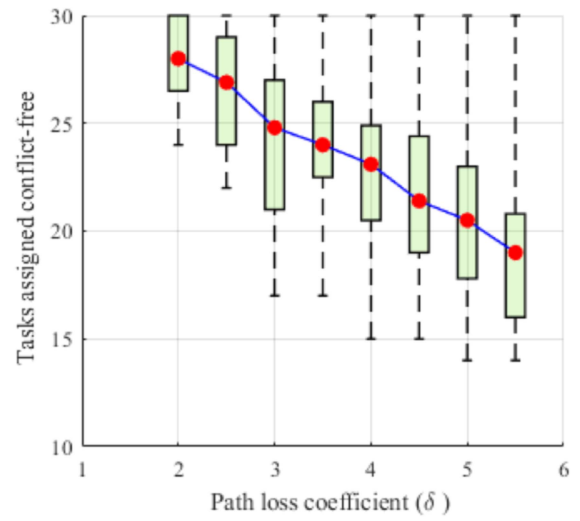


Fig. 12. Impact of path loss coefficients on Harmony DTA's conflict-free task assignment.

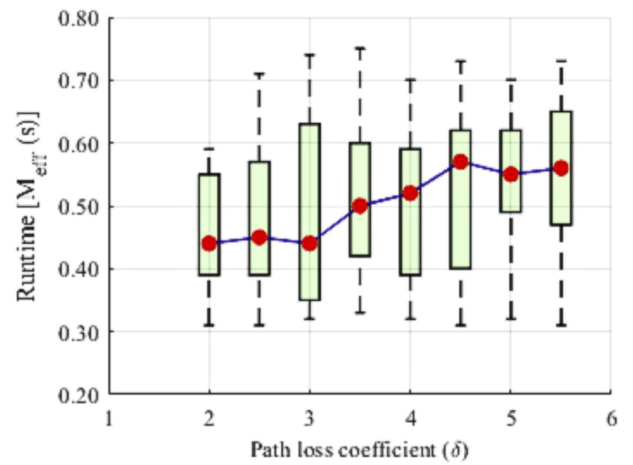


Fig. 13. Impact of path loss coefficients on Harmony DTA runtime (efficiency metric).

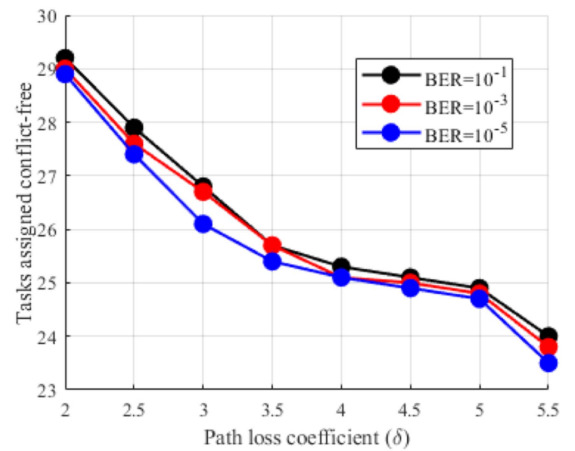


Fig. 14. Effect of BER on Harmony DTA's conflict-free task assignment performance.

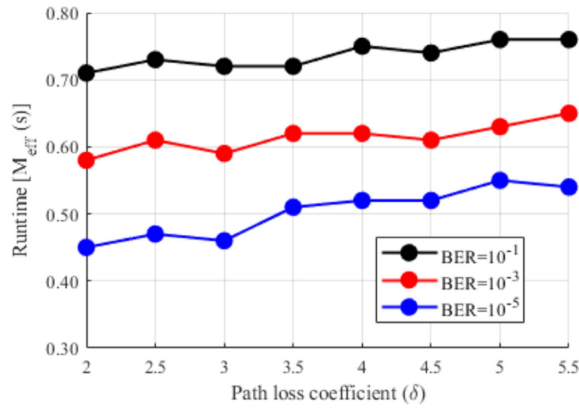


Fig. 15. Effect of BER on the execution time (efficiency metric) of Harmony DTA.

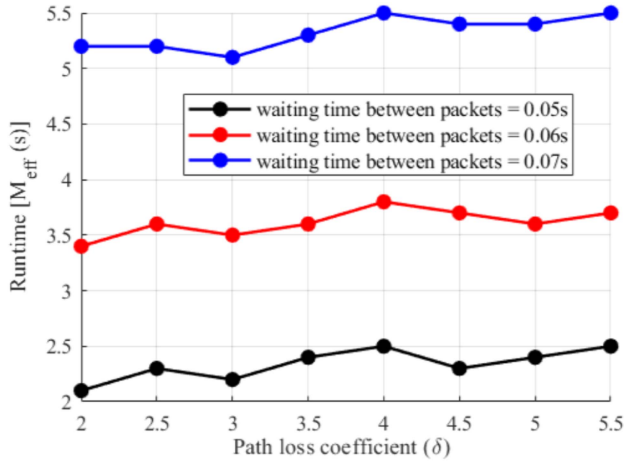


Fig. 16. Effect of maximum bit wait time on runtime (efficiency metric).

increase in the execution time required for the algorithm to complete the assignment.

As the third part of this section, simulation results aimed at analyzing the impact of the maximum bit waiting time on the performance of the Harmony DTA algorithm are presented. To examine the effect of maximum bit waiting time, the δ value was varied between 2 and 5.5 in increments of 0.5. For each δ value, simulations were conducted for maximum bit waiting times of 0.05, 0.06, and 0.07, with 100 simulations performed for each case. The average algorithm runtime (efficiency metric) values obtained from 100 simulations are shown in Fig. 16. From this figure, it can be observed that an increase in the maximum bit waiting time leads to an exponential rise in the algorithm's runtime. This increase is primarily attributed to the additional time required during the auction and gossip-based consensus phases.

4) *Comparative Analysis of Assignment Performance in Communication-Constrained Environments:* In this section, the performance of Harmony DTA is compared with CBBA, PI, and HIPC algorithms under a scenario with ten drones and 30 tasks. The comparison considers packet loss and BER factors for different communication coefficient

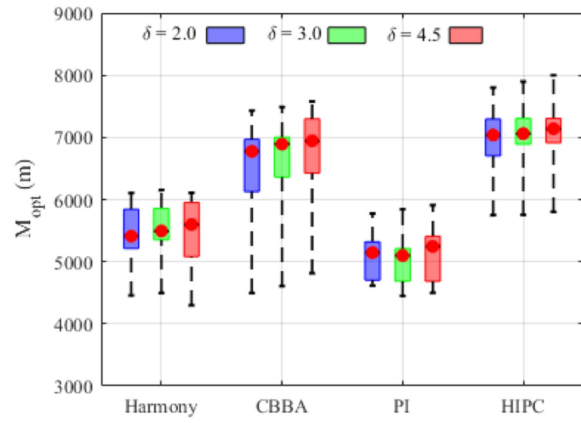


Fig. 17. Comparison of the optimality metric.

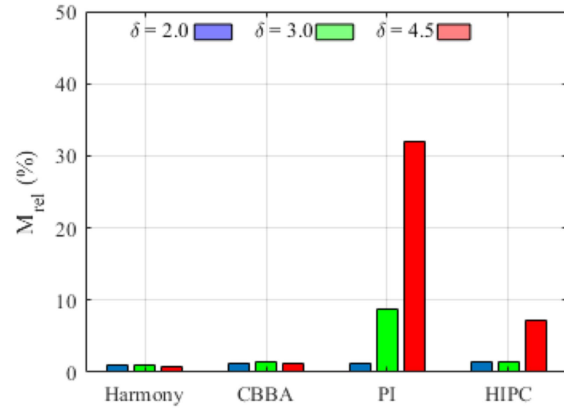


Fig. 18. Comparison of the reliability metric.

values, using the metrics of optimality, reliability, and communication load.

The results obtained for the optimality metric are presented in Fig. 17. The results used for the optimality analysis are derived from conflict-free assignments. In the comparison based on the optimality metric, it was observed that as communication became more challenging due to the path loss coefficient in various scenarios, there was no significant decline in the optimality of all algorithms. Since the PI algorithm focuses on global optimization, it achieved the best optimality in all cases. On the other hand, CBBA and HIPC, which rely more on local optimization strategies, performed below the PI algorithm in terms of optimality. Harmony, however, achieved better results than CBBA and HIPC due to its structure that not only focuses on local optimization but also considers the synergy between tasks. Nevertheless, its performance remained below that of the PI algorithm.

The results obtained for the reliability metric are presented in Fig. 18. The reliability metric of the PI algorithm shows that conflict rates remain below 10% at low path loss coefficients but approach 40% when the coefficient reaches 4.5. This is due to the additional messages required for the PI algorithm's min-max optimization, which increase the risk of conflicts. For HIPC, results indicate better performance than PI at high path loss coefficients but lower than CBBA

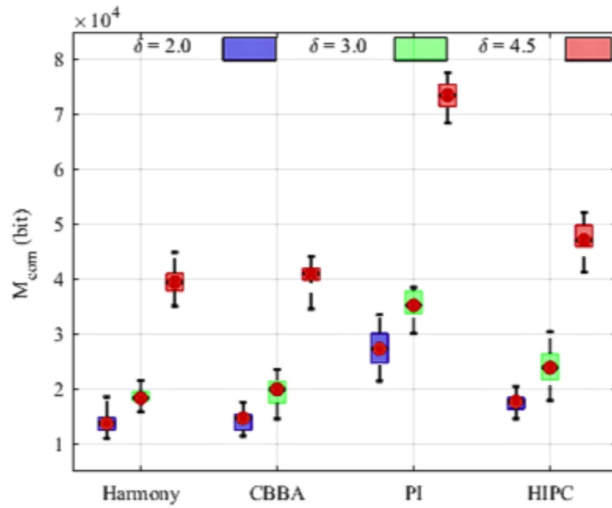


Fig. 19. Comparison of the communication load metric.

and Harmony. CBBA and Harmony maintain near-zero conflict rates in all scenarios, demonstrating high reliability in challenging communication conditions.

The results obtained for the communication load metric are presented in Fig. 19. The results can be summarized as follows: Communication disruptions increase the communication load for all algorithms. This is primarily due to the need for additional messages to compensate for the effects of packet loss, bit errors, and delays. Harmony demonstrates the best performance by solving the problem with smaller sized messages, thanks to its developed declarative ACL. In terms of communication load, CBBA has the second-lowest load after Harmony DTA, as CBBA packets have shorter bit lengths. HIPC also performs well in terms of communication load since it requires fewer iterations by sharing the AAVs' current positions to solve a global allocation on each local platform. On the other hand, while the PI algorithm provides better optimal solutions, it requires more communication and computational resources.

The Harmony DTA method significantly outperforms consensus-based bundle algorithms in terms of message size efficiency. The architecture of Harmony DTA includes an agent communication language in addition to the task allocation algorithm. Messages created using this language contain only the necessary information (e.g., sender/receiver IDs, task costs, and UCs). This compact message structure allows the problem to be solved with smaller sized messages.

In Harmony, assignments are made through messaging during a two-stage auction phase. In the gossip-based consensus phase, the lists of winning agents and winning bids are updated. However, in other algorithms, task allocations are indirectly carried out using these vectors, increasing the communication load. In contrast, in consensus-based bundle algorithms, agents communicate to achieve task assignments and consensus through N_t -dimensional vectors containing information, such as the winning agent, winning bid, and timestamp. This increases message size and

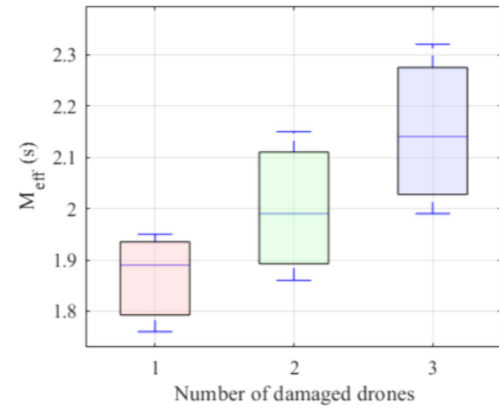


Fig. 20. Execution times for different numbers of drone failures (efficiency metric).

negatively impacts system performance, especially in networks with limited bandwidth, by raising communication costs.

H. Examination of Drone Damage Cases

In this section, the efficiency metric of Harmony DTA is analyzed for six drones and 15 tasks under varying numbers of drone failures. Results are presented in Fig. 20, obtained by running 100 simulations for each case. Harmony DTA achieves reassignment in an average of 1.89, 1.99, and 2.14 s for scenarios involving 1, 2, and 3 drone failures, respectively.

I. Performance Evaluation Through a Case Study

In Section V-D, the performance of Harmony DTA was evaluated in classical task assignment scenarios. In this section, however, Harmony DTA's application to modular drone swarms has been simulated to provide an additional contribution to literature. Modular drones are a type of drone that can divide into two, three, or six smaller drones mid-air, depending on mission requirements. These small drones, connected through a unifying body, can be easily assembled and disassembled on the ground, or autonomously split mid-air and move independently [44]. Although there are various studies on the flight performance and feasibility of modular drone swarms, which represent a relatively new technology, an extensive literature review revealed no studies addressing the distributed decision-making and task allocation problem in modular drone swarms.

The simulation begins with a swarm consisting of three divisible heterogeneous drones (D1, D2, and D3), and a total of five time-constrained tasks are required to be completed. While Type 1 drones can execute only task type 1, Type 2 drones are capable of executing only task type 2. It is further assumed that both drone types move at a maximum speed of 20 m/s. Fig. 21 shows the results of the initial task assignments obtained using the Harmony DTA algorithm. To improve the readability of the figures, the drone return routes to the base after completing their tasks have not been shown.

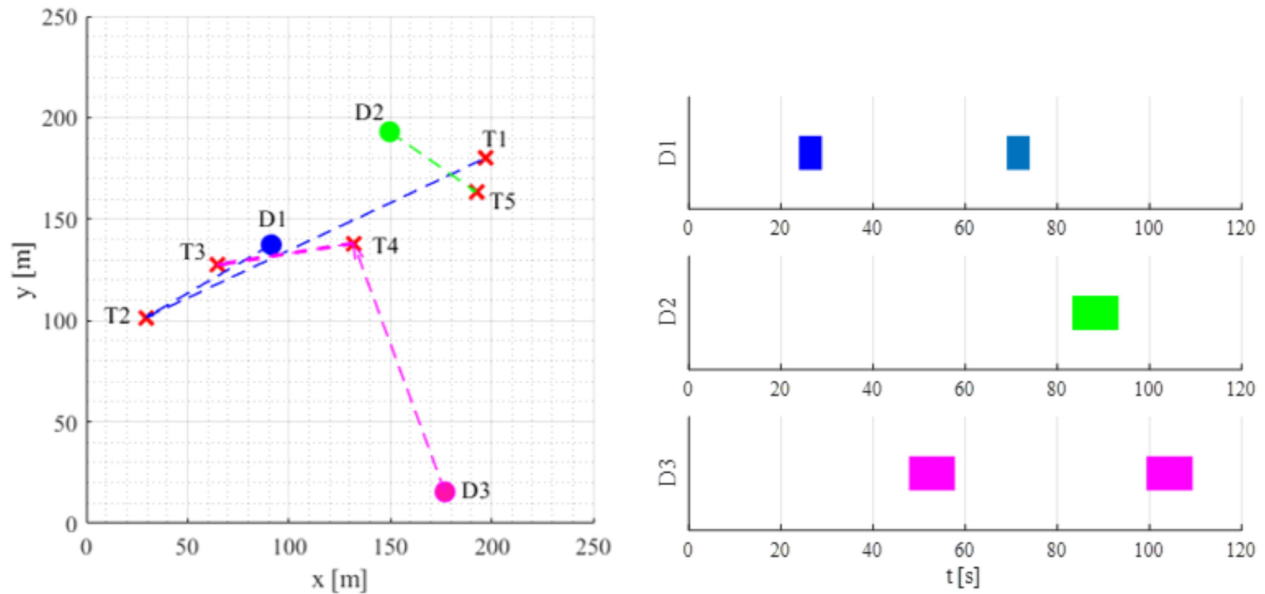


Fig. 21. Task assignment results and task execution schedule for the initial task list.

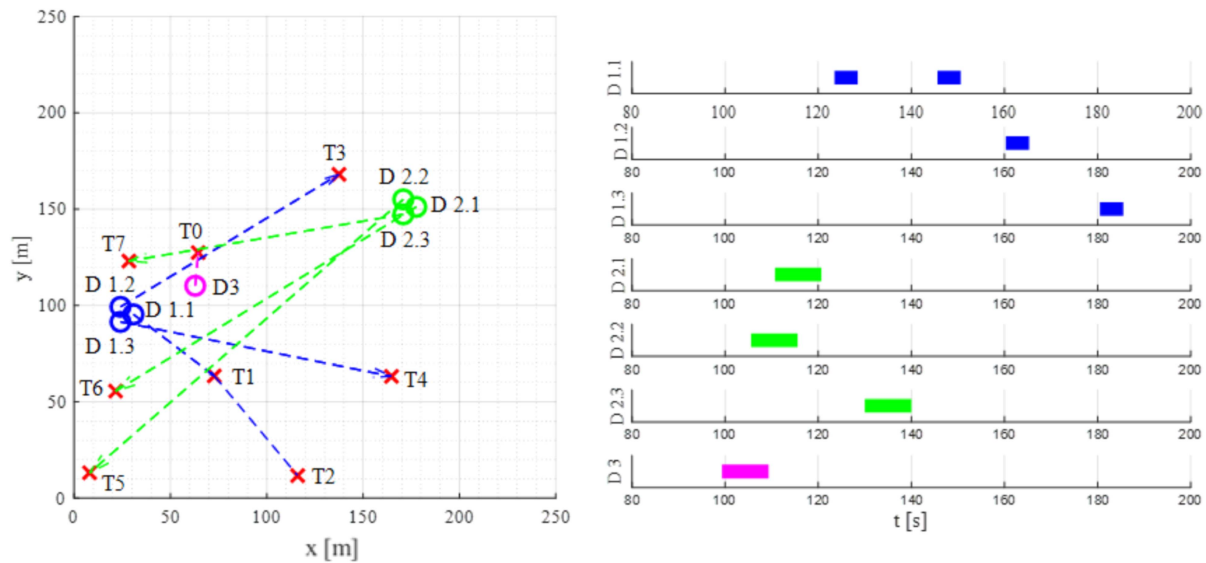


Fig. 22. Task assignment results and task execution schedule for the updated task list.

At the 93rd second of the simulation, a set of seven new tasks was announced to the drones by the ground control station. Since drone D3 had not yet completed its second task, T3 (which will be referred to as T0 for the remainder of the simulation), the total number of tasks that the swarm needed to complete increased to eight, including the newly announced tasks. To complete these tasks, drones D1 and D2 were each split into three units. As a result of the task assignment for the expanded swarm following the splitting, the drone routes and time schedules are illustrated in Fig. 22.

VI. CONCLUSION

The Harmony DTA algorithm introduced in this study provides a robust, efficient, and adaptable solution for decentralized task rescheduling in heterogeneous drone

swarms operating under dynamic and communication-constrained environments. The algorithm integrates a gossip-based consensus mechanism with a two-stage auction framework to achieve conflict free task assignments with a very small communication overhead. Its ability to adapt to dynamic conditions, including task additions, drone failures, and rogue agents, highlights its resilience and operational robustness. Furthermore, the modular design, encompassing submodules for health monitoring, trust-based security, and agent communication, enhances the system's reliability and overall mission success.

Simulation results validate the effectiveness of the Harmony DTA algorithm in minimizing total system cost, reducing task execution delays, and maintaining scalability across varying environmental complexities. Harmony DTA

achieves a good tradeoff between optimality and computational efficiency, and therefore is a competitive choice for real-world applications involving dynamic task assignment in decentralized drone swarms compared to benchmark algorithms.

Looking forward, future research should focus on expanding the capabilities of Harmony DTA to further address real-world complexities. Cost functions are expected to be generalized to include factors, such as energy consumption, drone capacities, and environmental conditions. Incorporating energy consumption into cost functions could enhance the swarm's operational efficiency by ensuring longer mission durations and optimal resource utilization. In addition, integrating environmental factors, such as wind direction, wind speed, and terrain variations, into cost calculations would enable the algorithm to better align with realistic operational scenarios, particularly in mission-critical applications.

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